

PROJECT TITLE

**SECURITY OPERATIONS CENTER
(SOC) MONITORING SYSTEM**

Submitted By

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ABSTRACT

In today's digital environment, organizations face a rapidly increasing number of cyber threats, making continuous security monitoring an essential component of cybersecurity. A **Security Operations Center (SOC)** plays a vital role in monitoring, detecting, analyzing, and responding to security incidents in real time. This project, titled "**Security Operations Center (SOC) Monitoring System**," demonstrates the practical implementation of SOC operations using industry-standard tools such as **Splunk Enterprise, Windows Event Viewer, and Wireshark**.

The project begins with the collection of security logs from Windows systems using Windows Event Viewer. These logs are then imported into Splunk for centralized log management and analysis. Using Splunk Search Processing Language (SPL), various security events are analyzed to identify failed login attempts, successful logins, suspicious IP addresses, unusual user activities, and other potential security threats. Interactive dashboards are created to visualize security events, making it easier for security analysts to monitor system activities and identify abnormal behavior.

To enhance security monitoring, real-time alerts are configured in Splunk to detect suspicious events such as multiple failed login attempts from the same user or IP address. These alerts enable timely identification of potential brute-force attacks and unauthorized access attempts. The project also includes the investigation of detected incidents by analyzing log evidence, identifying indicators of compromise, determining possible causes, assessing the impact, and recommending appropriate mitigation strategies.

Additionally, Wireshark is used to capture and analyze network traffic, providing insights into communication patterns and supporting network-level threat investigation. Combining host-based log analysis with network traffic monitoring offers a comprehensive understanding of system security and strengthens the incident investigation process.

The outcomes of this project include the development of an interactive SOC dashboard, generation of security alerts, preparation of an incident report, and creation of a detailed threat analysis report. The project demonstrates how centralized log management, continuous monitoring, and effective incident response contribute to strengthening an organization's cybersecurity posture.

INTRODUCTION

The rapid growth of information technology and the increasing dependence on digital systems have significantly improved communication, business operations, and data management. However, this digital transformation has also increased the number and sophistication of cyber threats such as malware attacks, phishing, ransomware, unauthorized access, insider threats, and brute-force attacks. These threats can compromise sensitive information, disrupt business operations, and cause significant financial and reputational damage. As a result, organizations require continuous monitoring and timely detection of security incidents to protect their information assets.

A **Security Operations Center (SOC)** is a centralized facility responsible for monitoring, detecting, analyzing, investigating, and responding to cybersecurity incidents. SOC analysts continuously observe security events generated by servers, workstations, network devices, applications, and security tools to identify suspicious activities and minimize security risks. By collecting and analyzing logs from multiple sources, a SOC helps organizations detect threats at an early stage and respond before they cause serious damage.

One of the most important components of a SOC is a **Security Information and Event Management (SIEM)** system. A SIEM platform collects logs from various systems, stores them in a centralized repository, and provides real-time analysis, visualization, alert generation, and incident investigation capabilities. In this project, **Splunk Enterprise** is used as the SIEM platform because of its powerful log indexing, searching, dashboard creation, and alerting features.

The project begins with collecting security logs from **Windows Event Viewer**, which records various system events such as successful logins, failed login attempts, user logouts, process creation, privilege assignments, and system errors. These logs are imported into Splunk, where they are analyzed using **Search Processing Language (SPL)** queries. The collected data helps identify important security events, including repeated failed login attempts, suspicious IP addresses, unusual user activities, and authentication anomalies.

In addition to host-based log analysis, **Wireshark** is used to capture and analyze network traffic. Network packet analysis provides valuable information about communication between devices, protocols in use, and suspicious network behavior. Combining host logs with network traffic analysis enables a more comprehensive investigation of security incidents.

An important feature of this project is the creation of an interactive **SOC Dashboard** that presents security events through charts, graphs, and statistical summaries. These visualizations help security analysts quickly identify abnormal activities and monitor the overall security posture of the system. Furthermore, Splunk alerts are configured to automatically notify administrators when predefined security conditions, such as multiple failed login attempts, are detected.

The project also demonstrates the complete incident response process, including event detection, investigation, evidence collection, threat analysis, impact assessment, and recommendation of mitigation strategies. An incident report and threat analysis report are prepared to document the findings and support future security improvements.

OBJECTIVES

The primary objective of this project is to understand the working of a **Security Operations Center (SOC)** by implementing a practical security monitoring environment using **Splunk Enterprise, Windows Event Viewer, and Wireshark**. The project focuses on collecting, monitoring, analyzing, and investigating security events to detect potential cyber threats and improve the overall security posture of an information system.

The specific objectives of this project are as follows:

1. **To understand the role and responsibilities of a Security Operations Center (SOC)** in monitoring, detecting, analyzing, and responding to cybersecurity incidents.
2. **To collect security logs** from Windows systems using Windows Event Viewer and understand the importance of event logs in digital security monitoring.
3. **To import and index system logs into Splunk Enterprise** for centralized log management and efficient security event analysis.
4. **To learn and use Splunk Search Processing Language (SPL)** for searching, filtering, and analyzing security events.
5. **To identify failed login attempts** by analyzing Windows Security Event Logs and detecting authentication-related security incidents.
6. **To detect suspicious IP addresses** by examining login patterns, source IP information, and repeated access attempts.
7. **To identify unusual system activities** by analyzing different security events, abnormal user behavior, and unexpected system operations.
8. **To create an interactive SOC dashboard** that visually represents security events through charts, tables, graphs, and statistical summaries for easier monitoring.
9. **To configure real-time security alerts** in Splunk for automatically detecting suspicious activities such as repeated failed login attempts and other critical events.
10. **To investigate detected security incidents** by collecting evidence, analyzing logs, identifying the root cause, assessing the impact, and documenting the findings.

SCOPE OF THE PROJECT

The scope of this project is to demonstrate the practical implementation of a **Security Operations Center (SOC) Monitoring System** by collecting, monitoring, analyzing, and investigating security events using industry-standard cybersecurity tools. The project provides hands-on experience with Security Information and Event Management (SIEM) technologies and introduces the fundamental processes involved in modern security operations.

The project focuses on collecting security logs from Windows systems through **Windows Event Viewer** and importing these logs into **Splunk Enterprise** for centralized storage and analysis. By using Splunk Search Processing Language (SPL), the project enables efficient searching, filtering, and visualization of security events. The analysis primarily covers authentication events such as successful logins, failed login attempts, user logouts, privilege assignments, and other important system activities.

Another important aspect of the project is the identification of suspicious activities, including repeated failed login attempts, abnormal authentication patterns, suspicious IP addresses, and unusual system behavior. These activities are analyzed to determine whether they indicate potential cybersecurity threats such as brute-force attacks or unauthorized access attempts.

The project also includes the design and implementation of an interactive **SOC Dashboard** using Splunk. The dashboard provides real-time visual representations of security events through charts, graphs, tables, and statistical summaries, enabling security analysts to monitor system activities more efficiently and identify anomalies quickly.

To improve threat detection capabilities, the project demonstrates the configuration of automated security alerts in Splunk. These alerts notify administrators whenever predefined security conditions, such as multiple failed login attempts or unusual authentication events, are detected. This automation supports faster incident detection and response.

In addition to host-based monitoring, the project utilizes **Wireshark** to perform basic network traffic analysis. Captured packets are examined to understand network communication patterns and identify potential network-related security issues that may complement the findings obtained from system logs.

The project further extends to the investigation of detected incidents by analyzing available evidence, identifying indicators of compromise (IoCs), determining the possible root causes, assessing the potential impact, and recommending appropriate mitigation strategies. The findings are documented through an Incident Report and a Threat Analysis Report, following standard cybersecurity reporting practices.

Although this project simulates the core functions of a Security Operations Center, it is intended for educational and laboratory purposes. It does not include advanced enterprise-level features such as large-scale log correlation, cloud security monitoring, endpoint detection and response (EDR), threat intelligence integration, or automated security orchestration. Nevertheless, the project provides a strong foundation for understanding SOC workflows and prepares students for practical applications in cybersecurity, digital forensics, and incident response.

SOFTWARE AND TOOLS USED

The **Security Operations Center (SOC) Monitoring System** was implemented using several industry-standard cybersecurity tools. Each tool performs a specific function in the collection, monitoring, analysis, and investigation of security events. The following software applications were used during the implementation of this project.

SPLUNK ENTERPRISE

Splunk Enterprise is a Security Information and Event Management (SIEM) platform used to collect, index, search, analyze, and visualize machine-generated data. It enables security analysts to monitor security events in real time, detect threats, and investigate incidents.

Purpose

- Import Windows Event Logs
- Analyze security events
- Create dashboards
- Generate alerts
- Investigate incidents

Key Features

- Centralized Log Management
- Dashboard Creation
- Alert Generation
- Real-Time Monitoring
- Search Processing Language (SPL)

WINDOWS EVENT VIEWER

Windows Event Viewer is a built-in Windows utility that records application, system, and security events generated by the operating system. It helps security analysts monitor login attempts, authentication failures, user activities, and system events.

Purpose

- Collect Security Logs
- Monitor Authentication Events
- Analyze Event IDs
- Detect Failed Login Attempts

Important Event IDs

Event ID	Description
4624	Successful Login

Event ID	Description
4625	Failed Login
4634	User Logout

WIRESHARK

Wireshark is a network protocol analyzer used to capture and inspect packets traveling through a network. It assists in identifying suspicious communication and supports network-level investigations.

Purpose

- Capture Network Packets
- Monitor Network Traffic
- Analyze Protocols
- Investigate Suspicious Communication

Key Features

- Live Packet Capture
- Protocol Analysis
- Packet Filtering
- Traffic Statistics
- Deep Packet Inspection

WINDOWS OPERATING SYSTEM

The Windows operating system serves as the environment where authentication events, security logs, and system activities are generated. These logs are later collected and analyzed in Splunk.

Purpose

- Generate Windows Event Logs
- Record Authentication Events
- Produce Security Logs
- Support Log Collection

SPLUNK SEARCH PROCESSING LANGUAGE (SPL)

Splunk Search Processing Language (SPL) is the query language used for searching, filtering, and analyzing indexed logs within Splunk.

It enables SOC analysts to detect suspicious activities and generate reports efficiently.

Example SPL Queries

- **Failed Login Detection**
index=wineventlog EventCode=4625
- **Successful Login Detection**
index=wineventlog EventCode=4624
- **Event Distribution**
index=wineventlog
| stats count by EventCode

SYSTEM ARCHITECTURE

The **System Architecture** of the Security Operations Center (SOC) Monitoring System illustrates the complete workflow of collecting, processing, analyzing, and monitoring security events. The architecture integrates multiple cybersecurity tools to provide centralized log management, real-time monitoring, threat detection, incident investigation, and security reporting.

In this project, logs generated by the Windows operating system are collected using **Windows Event Viewer**, while network traffic is captured using **Wireshark**. The collected logs and network data are then analyzed using **Splunk Enterprise**, where Security Operations Center (SOC) analysts perform log analysis, create dashboards, configure alerts, and investigate security incidents.

The architecture follows a structured flow from **data generation to incident response**, enabling efficient monitoring of system activities and early detection of potential security threats.

SYSTEM ARCHITECTURE DIAGRAM

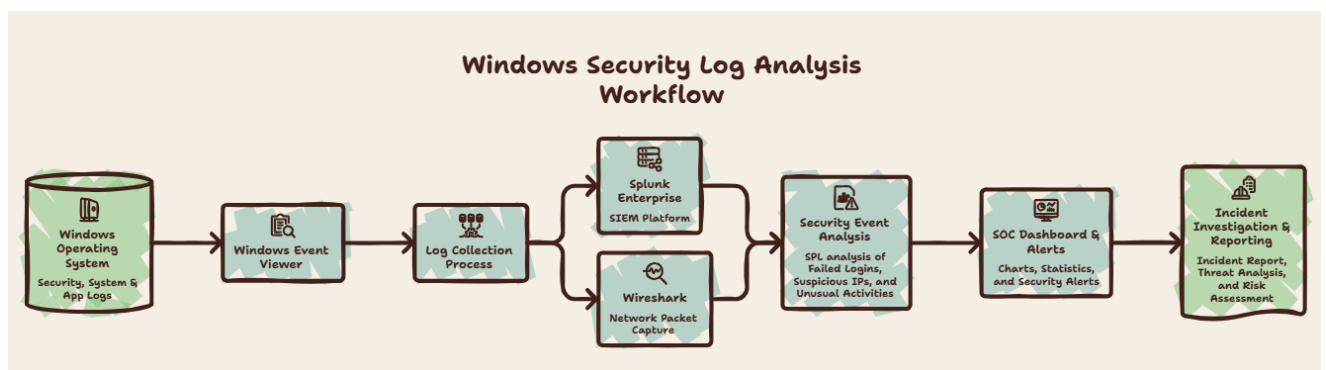


Figure 1

COMPONENTS OF THE SYSTEM

1. Windows Operating System

The Windows operating system acts as the primary source of security events. User logins, authentication attempts, privilege assignments, process creation, and other system activities generate logs that are stored in Windows Event Viewer.

2. Windows Event Viewer

Windows Event Viewer collects and stores system-generated logs. These logs contain valuable information related to authentication, user activities, system errors, and security events. The collected logs are imported into Splunk for centralized monitoring and analysis.

3. Wireshark

Wireshark captures live network packets exchanged between devices. During incident investigations, packet analysis helps identify suspicious communication, abnormal traffic patterns, and potential network-based attacks.

4. Splunk Enterprise

Splunk Enterprise serves as the central SIEM platform. It indexes collected logs, allows analysts to perform searches using SPL, creates dashboards, generates alerts, and supports security investigations.

5. Security Analysis

Security analysts use Splunk Search Processing Language (SPL) to detect failed login attempts, suspicious IP addresses, abnormal authentication patterns, and unusual system activities.

6. Dashboard and Alert System

The analyzed data is presented through interactive dashboards containing tables, charts, and graphs. Automated alerts notify analysts whenever suspicious activities exceed predefined thresholds.

7. Incident Investigation

Detected incidents are investigated using both Splunk logs and Wireshark packet captures. The investigation identifies the root cause, evaluates the severity, and recommends mitigation measures.

WORKFLOW OF THE SOC MONITORING SYSTEM

The overall workflow of the project consists of the following stages:

1. Windows generates security events during normal system operation.
2. Windows Event Viewer records all authentication and security logs.
3. Wireshark captures network traffic for packet-level analysis.
4. Security logs are imported into Splunk Enterprise.
5. Splunk indexes the collected log data.
6. SPL queries analyze security events and identify suspicious activities.
7. Dashboards visualize security statistics and trends.
8. Alerts notify the SOC analyst when suspicious events are detected.
9. Security incidents are investigated using logs and network packets.
10. Incident reports and threat analysis reports are prepared based on the investigation.

LOG COLLECTION PROCESS

Log collection is the first and one of the most important phases of a **Security Operations Center (SOC)**. Every activity performed on a computer or network generates digital records called **logs**. These logs contain valuable information about user authentication, system events, application activities, network communication, and security incidents. By collecting and analyzing these logs, SOC analysts can detect suspicious activities, investigate incidents, and respond to potential cyber threats.

In this project, log collection was performed using **Windows Event Viewer** to gather Windows security logs and **Wireshark** to capture network traffic. The collected data was then imported into **Splunk Enterprise** for centralized monitoring and analysis.

OBJECTIVES OF LOG COLLECTION

The primary objectives of the log collection process are:

- Collect security-related events generated by the operating system.
- Record successful and failed authentication attempts.
- Monitor user activities and system events.
- Capture network traffic for packet-level analysis.
- Store logs in a centralized SIEM platform.
- Provide data for threat detection and incident investigation.

LOG COLLECTION FROM WINDOWS EVENT VIEWER

Windows Event Viewer is a built-in utility that stores system, application, and security logs generated by the Windows operating system. Security logs contain authentication events, privilege assignments, process creation events, and other activities useful for cybersecurity monitoring.

Steps to Collect Windows Security Logs

Step 1: Press **Windows + R** to open the Run dialog box.

Step 2: Type the following command and press **Enter**: eventvwr.msc

Screenshot

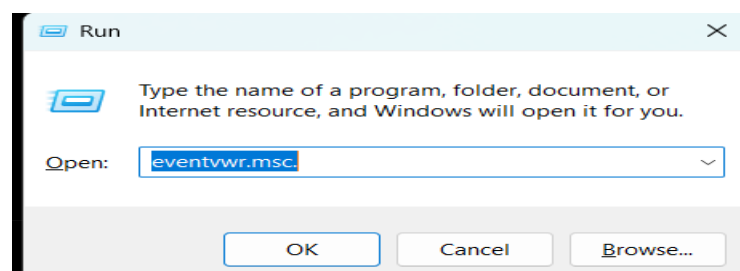


Figure 2: Opening Windows Event Viewer

Step 3: In the left navigation pane, expand:

Windows Logs



Security

Screenshot 3:

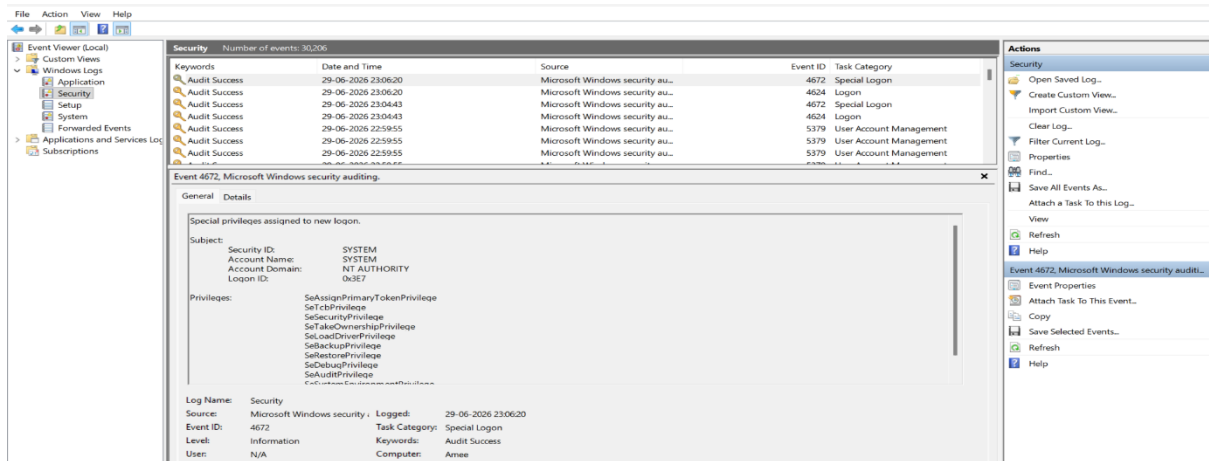


Figure 3: Windows Security Logs

Step 4: Review the security events generated by the operating system. The project primarily focused on the following Event IDs:

Event ID	Description
4624	Successful Login
4625	Failed Login
4634	User Logoff

Screenshot

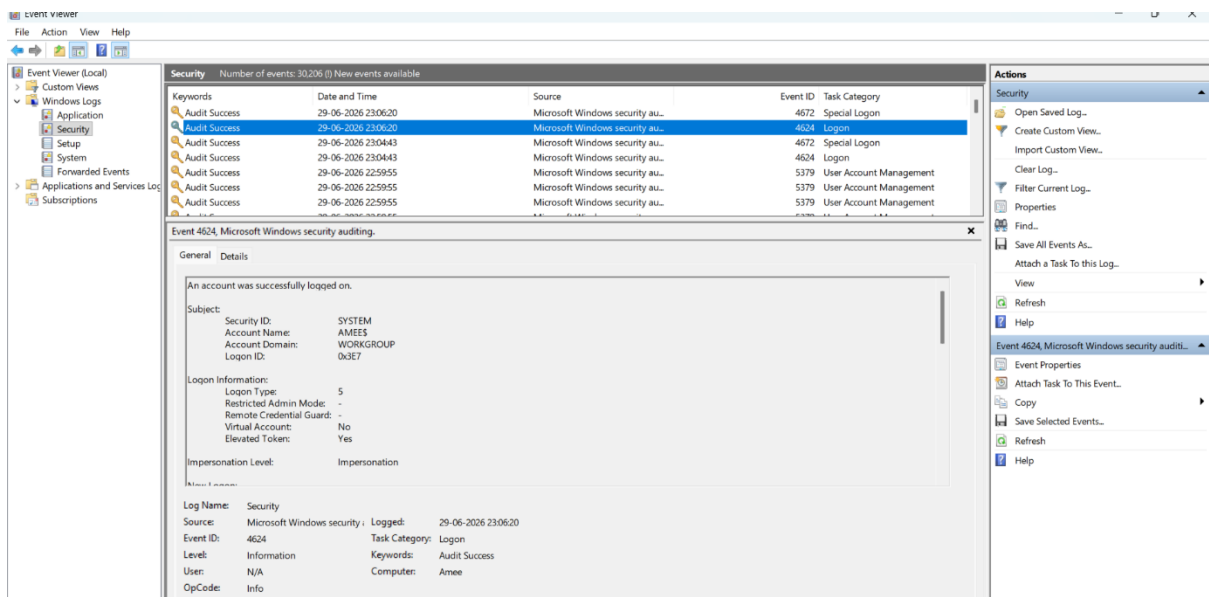


Figure 4: Windows Security EventID: 4624

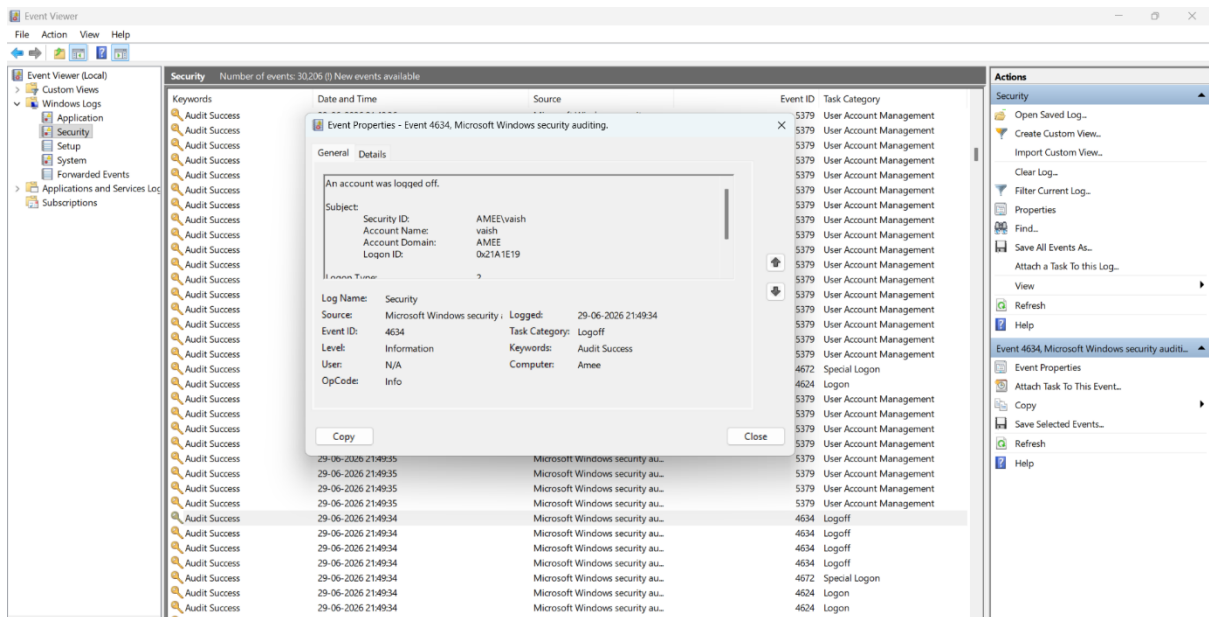


Figure 5: Windows Security EventID: 4634

NETWORK PACKET CAPTURE USING WIRESHARK

While Windows Event Viewer records host-based events, Wireshark captures packets transmitted across the network. Packet analysis helps detect suspicious communication, unknown hosts, protocol anomalies, and unusual network behavior.

Steps to Capture Network Traffic

Step 1: Launch Wireshark.

Step 2: Select the active network interface (Wi-Fi or Ethernet).

Screenshot:

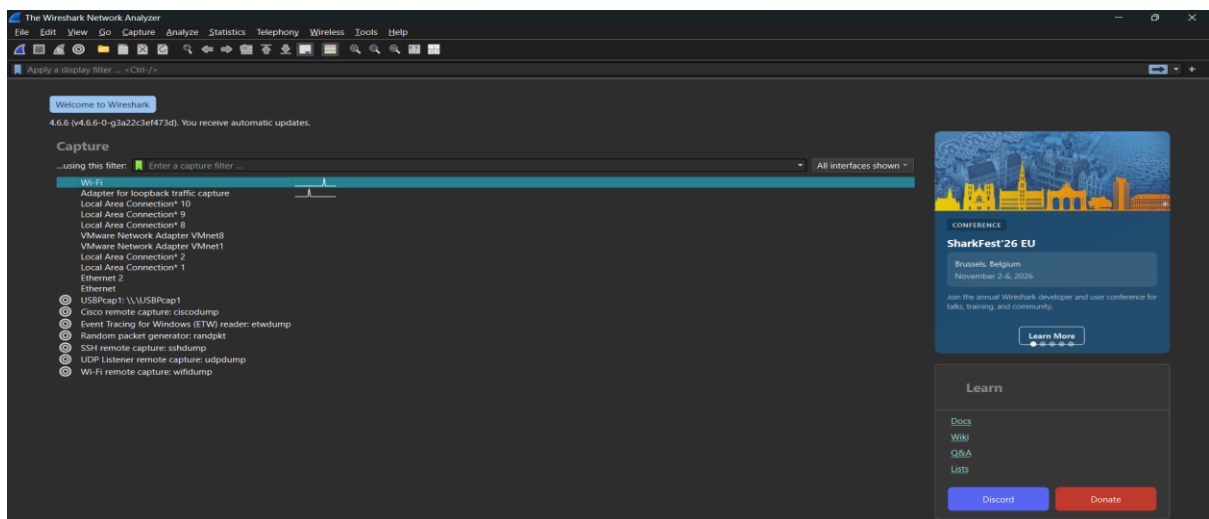


Figure 6: Selecting the Network Interface

Step 3: Click the Start Capture button.

Wireshark begins capturing live packets from the selected interface.

Screenshot

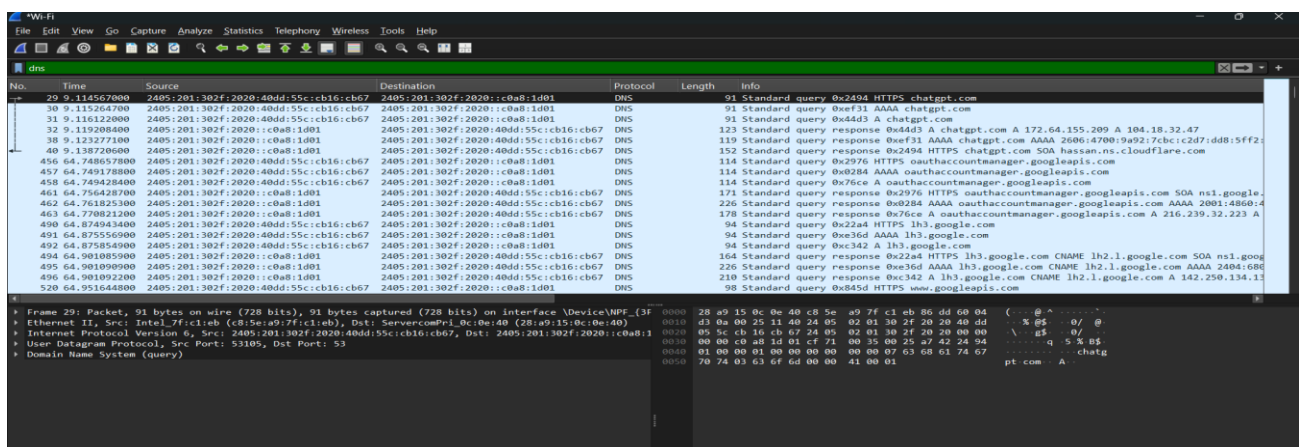


Figure 7: Live Packet Capture

Step 4: Apply protocol filters to analyze specific traffic.

Examples:

- dns

Screenshot

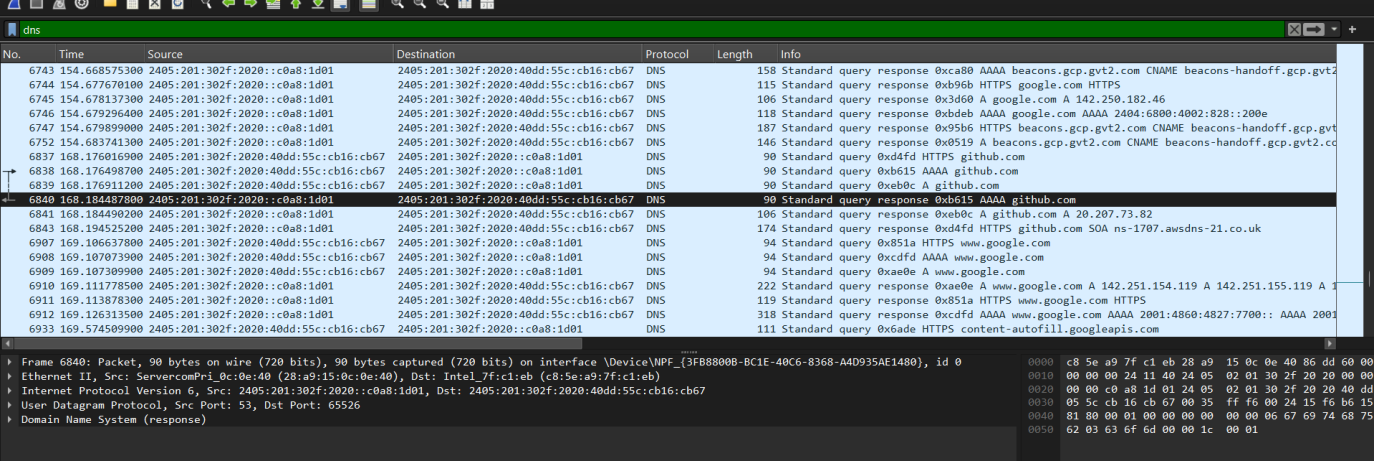


Figure 8:Applying Packet Filters

The following data was collected during the project:

Field	Value
Packet No.	6840
Time	168.184478800
Source	2405:201:302f:2020::c0a8:1d01
Destination	2405:201:302f:2020:40dd:55cc:cb16:cb67
Protocol	DNS
Length	90 bytes
Transaction ID	0xb615
Query Type	AAAA (IPv6 Address)
Domain Name	github.com
Packet Description	Standard DNS query response for the IPv6 (AAAA) record of github.com.

IMPORTANCE OF LOG COLLECTION

Effective log collection provides several advantages:

- Detects unauthorized login attempts.
- Supports incident investigation.
- Provides evidence for forensic analysis.
- Identifies abnormal user activities.
- Improves threat detection capabilities.
- Enables centralized security monitoring.
- Supports compliance and auditing.

IMPORTING LOGS INTO SPLUNK

After collecting security logs from the Windows operating system, the next step is to import these logs into **Splunk Enterprise**. Splunk acts as a centralized **Security Information and Event Management (SIEM)** platform that stores, indexes, and analyzes machine-generated data from various sources.

In this project, Windows Event Logs were imported into Splunk using **Local Event Log Data Inputs**. Once the logs were indexed, they became available for searching, dashboard creation, alert generation, and incident investigation.

OBJECTIVES

The objectives of importing logs into Splunk are:

- Centralize security logs in one platform.
- Enable real-time monitoring of system activities.
- Index log data for fast searching.
- Perform security analysis using SPL queries.
- Support dashboard creation and alert generation.
- Facilitate incident investigation.

STEPS TO IMPORT WINDOWS EVENT LOGS INTO SPLUNK

Step 1: Open Splunk Enterprise

1. Start the Splunk service .
2. Open a web browser.
3. Enter: `http://localhost:8000`
4. Log in using the administrator credentials.

Screenshot

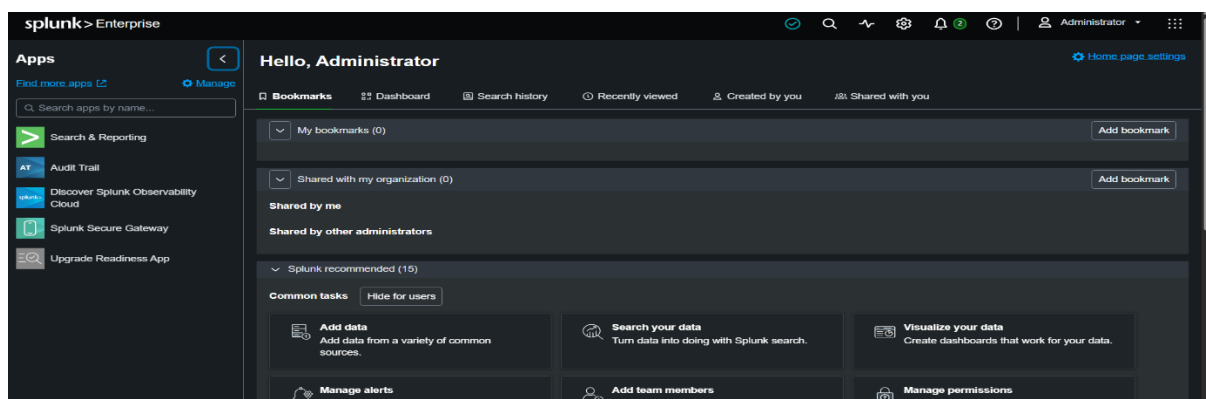


Figure 9: Splunk home Page

Step 2: Open Settings

After logging in:

1. Click **Settings** from the top menu.
2. Select **Data Inputs**.

This section allows Splunk to receive data from different sources.

Step 3: Configure Local Event Logs

- Under **Data Inputs**, select:
- **Local Event Logs**
- Click **New Local Event Log** (or **Add New**, depending on the Splunk version).

Step 4: Select Event Log Type

Choose the Windows log categories to monitor.

For this project, the following logs were selected:

- Security
- Setup
- System
- Application (optional)

The **Security** log was the primary source used for authentication monitoring.

Step 5: Configure Input Settings

Provide the required configuration details.

Click **Next** and then **Submit**.

Screenshot

The screenshot displays the 'Add Data' configuration wizard in Splunk. At the top, a progress bar indicates the current step is 'Review', with previous steps 'Select Source' and 'Input Settings' completed, and 'Done' remaining. Navigation buttons include '< Back' and a green 'Submit >' button. The main content area is titled 'Review' and contains the following configuration details:

Input Type	Windows Event Logs
Event Logs	Security Setup System Application
App Context	search
Host	Amee
Index	default

Figure 10: Input Configuration

Step 6: Verify Imported Logs

Open **Search & Reporting**. Run the following query: `source="WinEventLog:*" host="Amees"`

This query displays all indexed events available in Splunk.

Screenshot

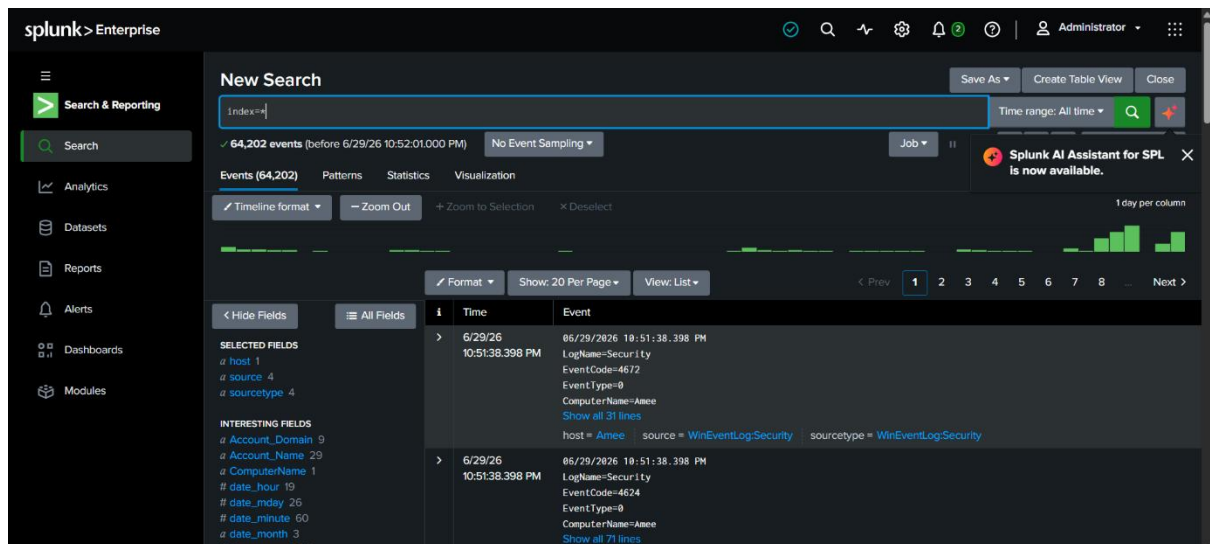


Figure 11: Verifying Imported Logs

Step 7: Verify Windows Security Events

To verify that Windows Security logs were imported successfully, execute: `source="WinEventLog:*" host="Amees"`

The search results should display authentication and security events collected from Windows Event Viewer.

Screenshot

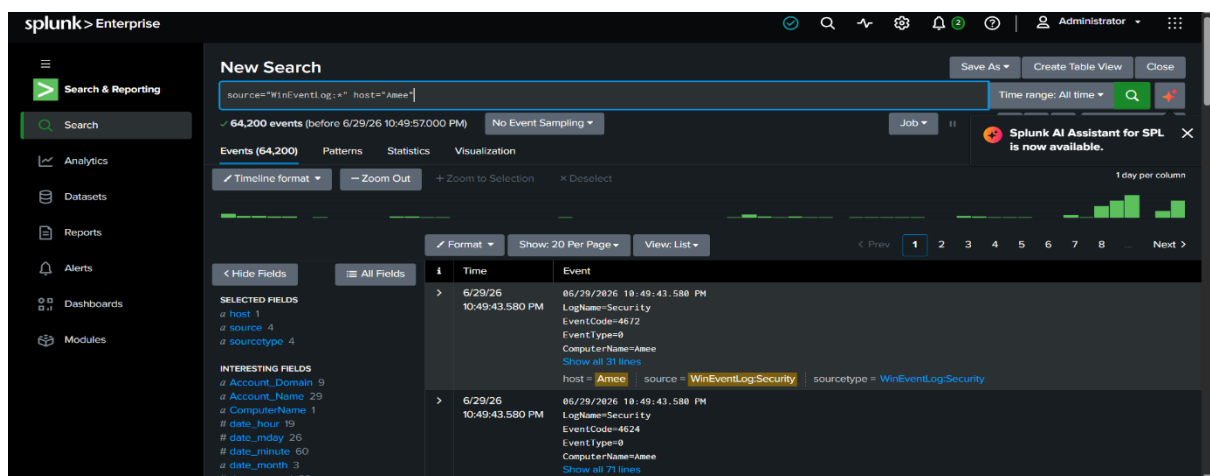


Figure 12: Windows Security Events in Splunk

SPLUNK SEARCH PROCESSING LANGUAGE (SPL) QUERIES

After successfully importing Windows security logs into Splunk Enterprise, the next step is to analyze the collected data using **Splunk Search Processing Language (SPL)**. SPL is Splunk's powerful query language that enables security analysts to search, filter, correlate, and analyze machine-generated data efficiently. In a Security Operations Center (SOC), SPL is extensively used to detect security incidents, monitor authentication activities, identify suspicious behavior, generate statistical reports, and support incident investigations. In this project, several SPL queries were executed to analyze Windows Security Logs and prepare dashboards and reports.

BASIC SPL QUERY

The following query displays all events stored in Splunk.

SPL Query: `index=*`

Purpose

- Displays all indexed events.
- Verifies that data has been successfully imported.
- Confirms that Splunk is receiving logs.

Screenshot

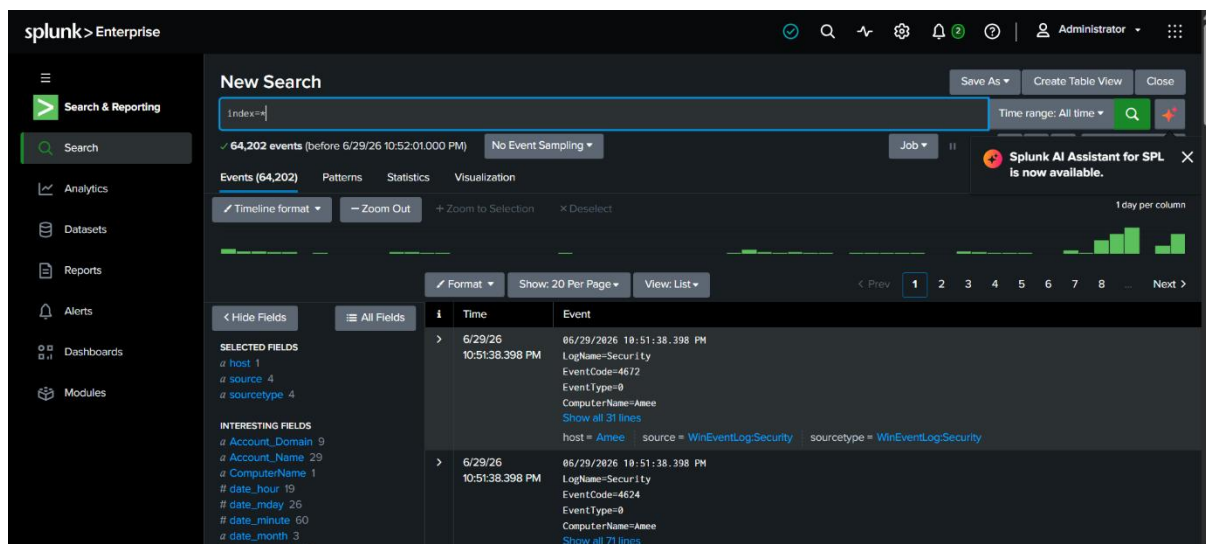


Figure 13: Displaying All Indexed Events

WINDOWS SECURITY LOG QUERY

The following query displays all Windows Security Events.

SPL Query:`index=main sourcetype="WinEventLog:Security"`

Purpose

- Displays all security events.
- Shows authentication logs.
- Provides data for further analysis.

Screenshot

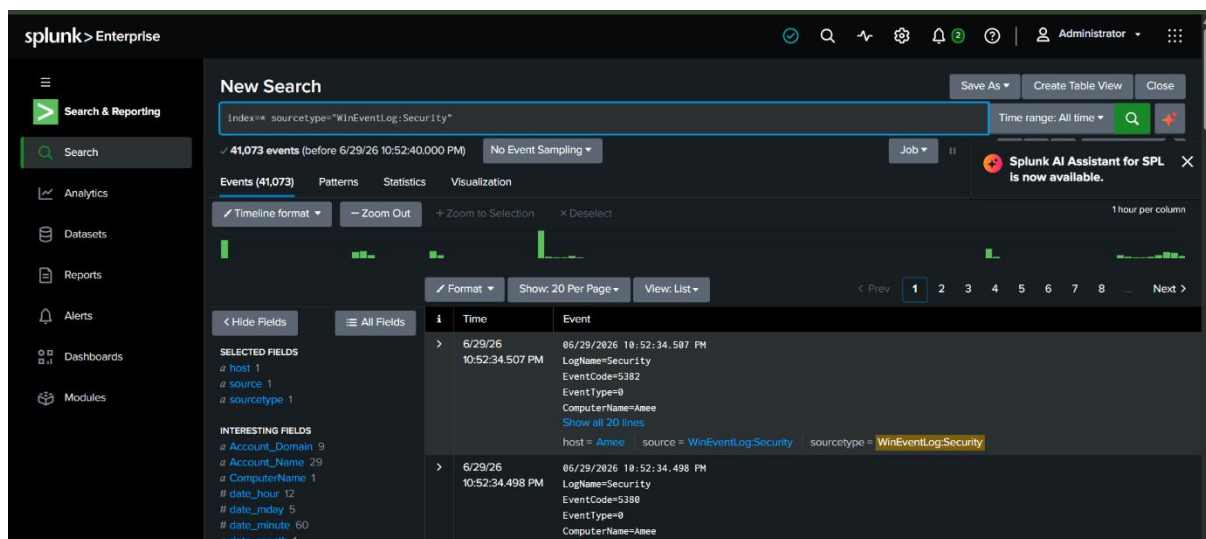


Figure 14: Windows Security Logs

FAILED LOGIN DETECTION QUERY

The following SPL query identifies all failed login attempts.

SPL Query:`index=wineventlog EventCode=4625`

Purpose

- Detects failed authentication attempts.
- Identifies brute-force attack attempts.
- Monitors unauthorized login activities.

Expected Output

The results display:

- Time
- Username
- Host
- Event ID (4625)

- Failure Reason

Screenshot

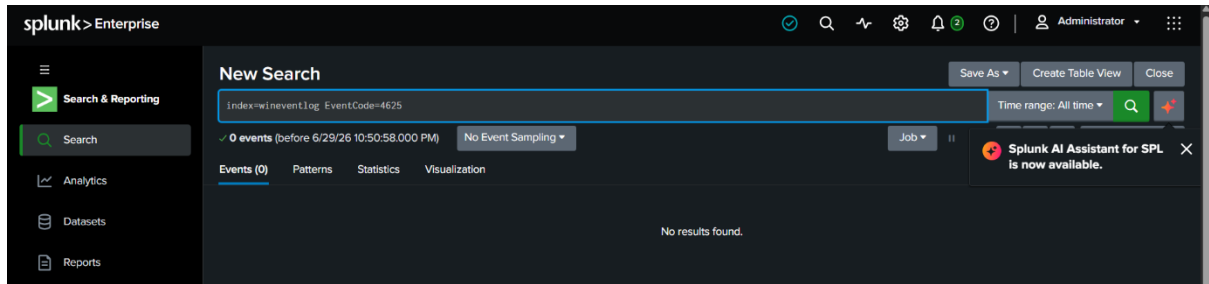


Figure 15: Failed Login Events

SUCCESSFUL LOGIN QUERY

This query retrieves all successful login events.

SPL Query:index=wineventlog EventCode=4624

Purpose

- Displays successful logins.
- Monitors user authentication.
- Compares successful and failed login attempts.

Screenshot

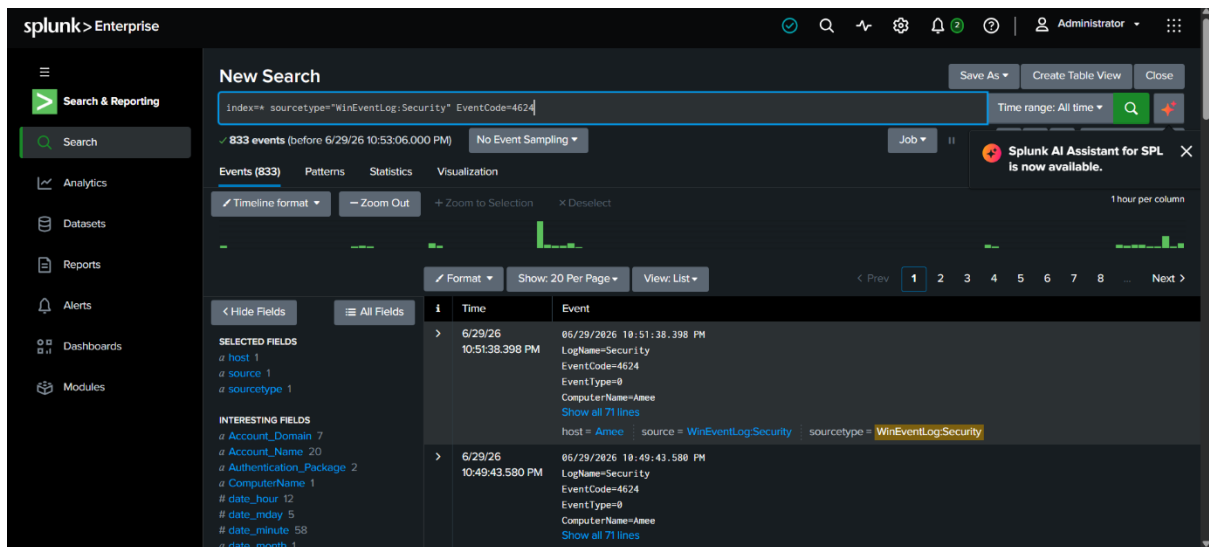


Figure 16: Successful Login Events

USER LOGOUT QUERY

The following query displays user logout events.

SPL Query:index=wineventlog EventCode=4634

Purpose

- Monitors user logout activities.
- Tracks session termination.
- Assists in user activity analysis.

Screenshot

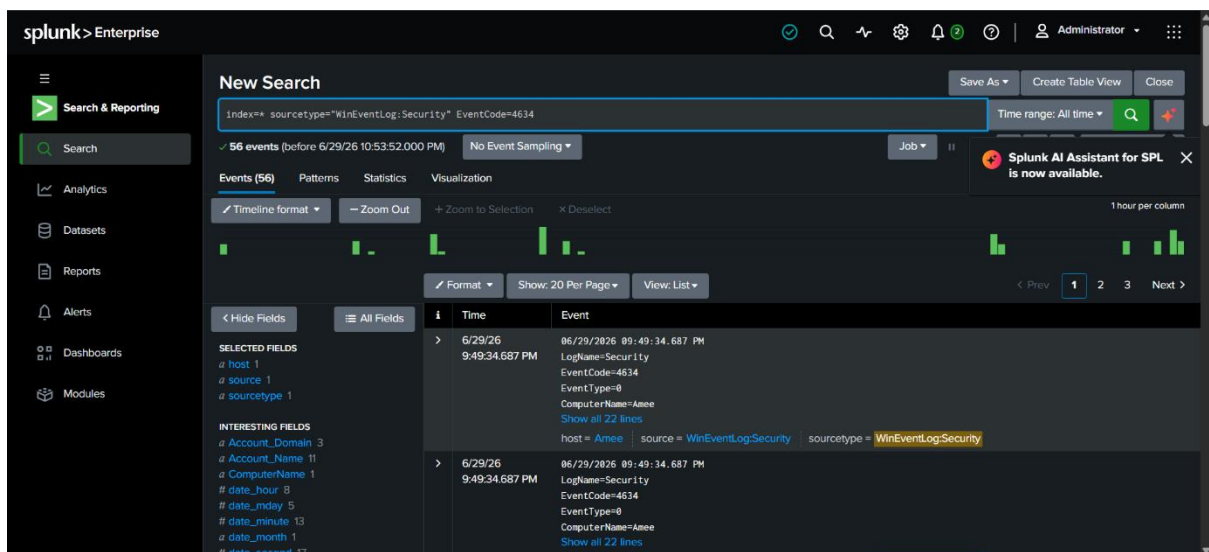


Figure 17: User Logout Events

EVENT DISTRIBUTION QUERY

SPL Query

`index=* sourcetype="WinEventLog:Security"`

`| stats count by EventCode`

Purpose

- Counts the occurrence of each Event ID.
- Identifies frequently occurring security events.
- Supports dashboard visualization.

Screenshot

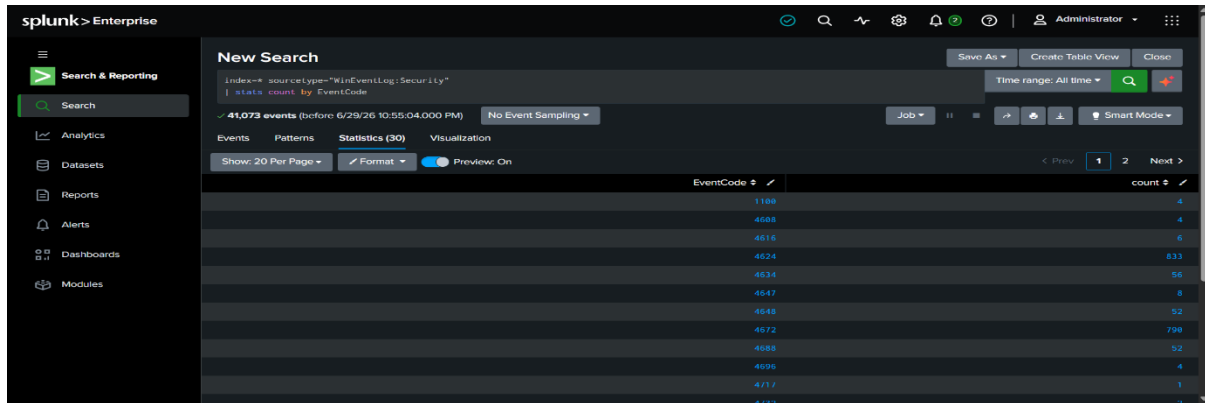


Figure 18: Event Distribution

TOP HOSTS QUERY

This query identifies systems generating the highest number of events.

SPL Query

```
index=* sourcetype="WinEventLog:Security"
```

```
| top host
```

Purpose

- Displays the most active hosts.
- Helps identify systems with unusual activity.
- Supports infrastructure monitoring.

Screenshot

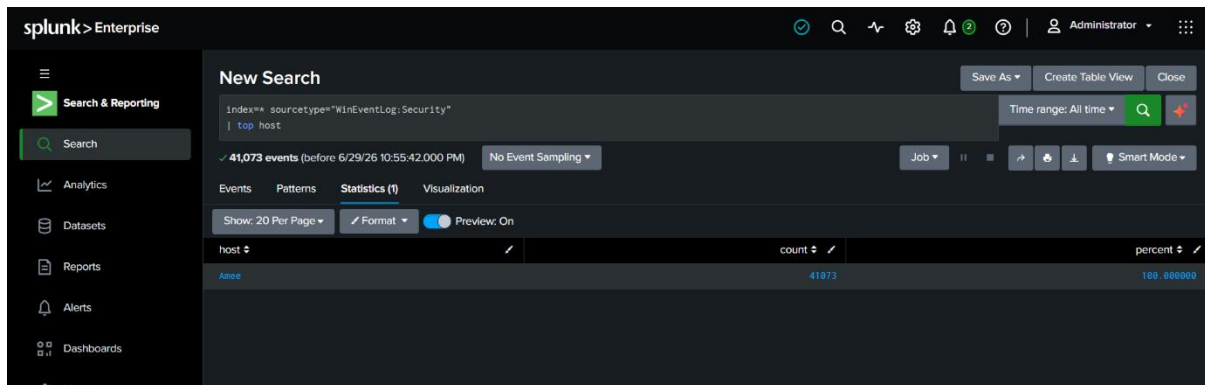


Figure 19: Top Hosts

LOGIN ACTIVITY OVER TIME

This query displays authentication events over time.

SPL Query

```
index=* sourcetype="WinEventLog:Security"
```

```
(EventCode=4624 OR EventCode=4625)
```

```
| timechart count by EventCode
```

Purpose

- Visualizes login trends.
- Detects spikes in failed login attempts.
- Supports security monitoring dashboards.

Screenshot

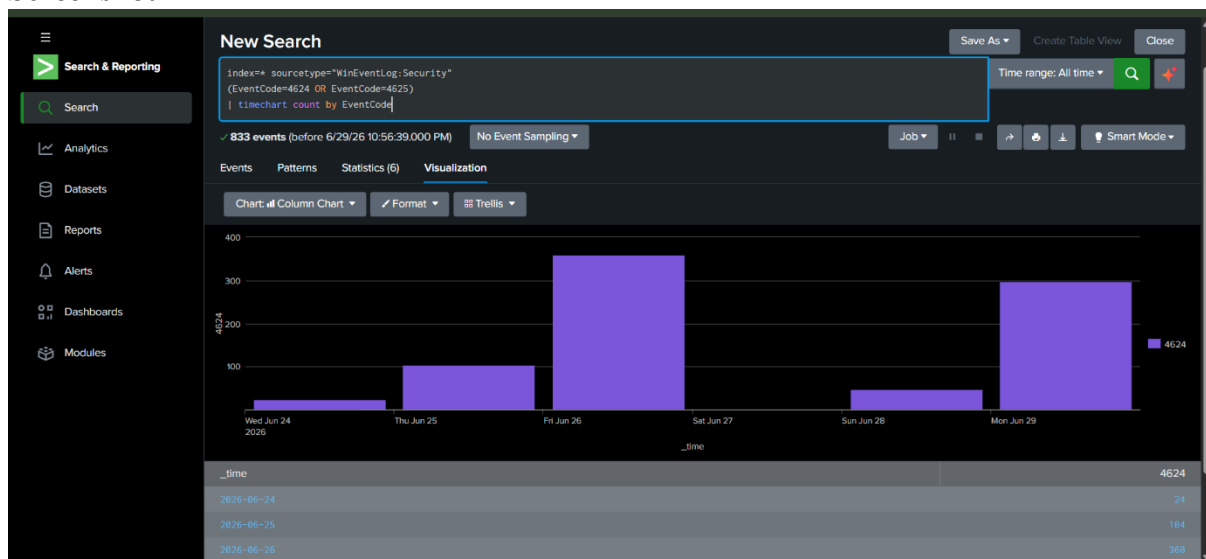


Figure 20: Login Activity Timeline

IMPORTANCE OF SPL QUERIES

Splunk Search Processing Language plays a vital role in SOC operations because it enables analysts to:

- Search large volumes of log data quickly.
- Detect suspicious authentication attempts.
- Analyze user activities.
- Correlate security events.
- Create dashboards and visualizations.
- Configure automated alerts.
- Investigate security incidents.
- Generate reports for management.

DETECTION OF FAILED LOGIN ATTEMPTS

Failed login attempts are one of the most common indicators of unauthorized access attempts. Attackers often try multiple username and password combinations to gain access to a system, a technique known as a **brute-force attack** or **password guessing attack**. Monitoring failed login events helps Security Operations Center (SOC) analysts identify suspicious authentication activities before they become successful attacks.

In this project, failed login attempts were detected by analyzing Windows Security Event Logs in Splunk using **Event ID 4625**. The search results were examined to identify the time of the event, affected user account, source host, and the reason for login failure.

SPL QUERY FOR FAILED LOGIN DETECTION

The following SPL query was used to retrieve failed login events:

```
index=wineventlog EventCode=4625
```

Purpose

- Retrieves all failed login events.
- Displays unsuccessful authentication attempts.
- Assists in identifying suspicious login behavior.

Screenshot

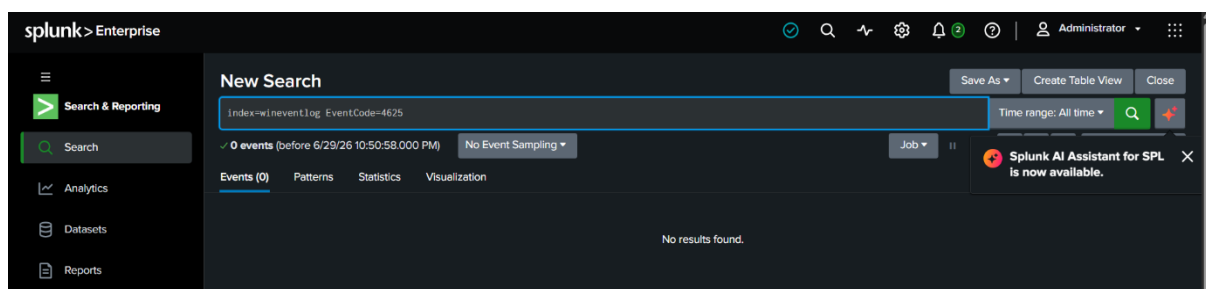


Figure 21: Failed Login Events in Splunk

ANALYSIS OF FAILED LOGIN EVENTS

The search was executed successfully in Splunk Enterprise using the specified query. However, the search returned **0 events**, indicating that no failed login attempts (Event Code 4625) were found in the selected index during the chosen time range.

The "No results found" message suggests one of the following:

- No failed login attempts occurred during the selected time period.

- Windows Security logs containing Event ID 4625 were not available in the configured index.
- The event logs may not have been ingested into Splunk.
- The selected time range did not include any failed authentication events.

Although no failed login events were detected, performing this search confirms that the monitoring process is correctly configured and can be used to detect failed authentication events whenever they occur.

OBSERVATION TABLE

<i>Parameter</i>	<i>Observation</i>
<i>Search Query</i>	<i>index=wineventlog EventCode=4625</i>
<i>Event Code</i>	<i>4625</i>
<i>Events Found</i>	<i>0</i>
<i>Search Result</i>	<i>No results found</i>
<i>Time Range</i>	<i>All Time</i>
<i>Status</i>	<i>No failed login events detected</i>

INDICATORS OF SUSPICIOUS ACTIVITY

Repeated failed login attempts may indicate:

- Brute-force password attacks.
- Password guessing attempts.
- Unauthorized access attempts.
- Use of invalid or disabled user accounts.
- Misconfigured applications repeatedly attempting authentication.

SOC ANALYST INVESTIGATION

In this project, no further investigation was required because the search did not return any failed login events.

IDENTIFICATION OF SUSPICIOUS IP ADDRESSES

Identifying suspicious IP addresses is an important activity in a **Security Operations Center (SOC)** because attackers often attempt unauthorized access from the same source IP address. By monitoring authentication logs, SOC analysts can identify repeated login attempts, detect brute-force attacks, and investigate suspicious network activity.

In this project, **Splunk Enterprise** was used to analyze Windows Security Logs for failed login events (**Event Code 4625**) and to identify the **Source Network Address** associated with those events. The search was performed to determine whether any IP addresses were responsible for repeated failed authentication attempts.

SPL QUERY FOR SOURCE IP ADDRESS ANALYSIS

The following SPL query was used to search for failed login attempts and group them according to the source network address:

```
index=* sourcetype="WinEventLog:Security" EventCode=4625  
| stats count by Source_Network_Address  
| sort -count
```

Purpose

- Searches Windows Security logs for **Event Code 4625**.
- Groups failed login events by **Source_Network_Address**.
- Counts the number of failed login attempts from each IP address.
- Helps identify IP addresses responsible for repeated authentication failures.

Screenshot

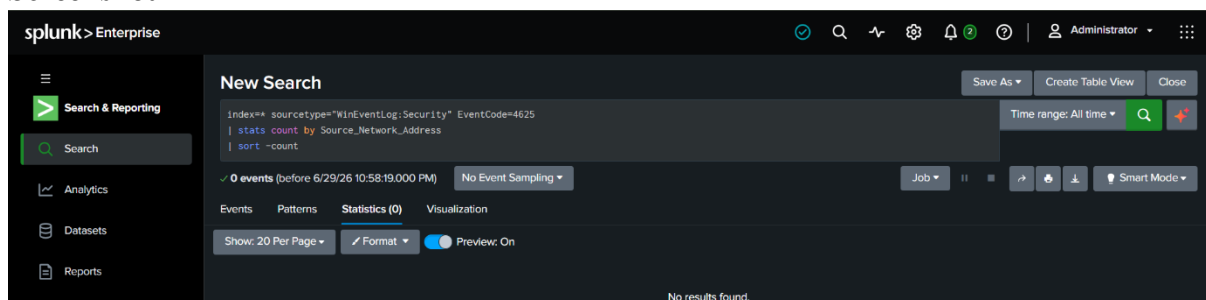


Figure 22: Failed Login Attempts by Source Network Address

ANALYSIS OF SEARCH RESULTS

The SPL query was executed successfully in Splunk Enterprise. However, the search returned **0 events**, indicating that no failed login events containing the **Source_Network_Address** field were found during the selected time range.

The "**No results found**" message indicates one or more of the following:

- No failed login attempts occurred during the selected period.
- Windows Security logs containing Event Code 4625 were not available in the Splunk index.
- Source IP address information was not present in the collected event logs.
- The selected time range did not contain any matching authentication events.

As a result, no suspicious IP addresses were identified during this analysis.

OBSERVATION TABLE

Parameter	Observation
Search Query	index=* sourcetype="WinEventLog:Security" EventCode=4625 stats count by Source_Network_Address
Event Code	4625
Source Field	Source_Network_Address
Events Found	0
Suspicious IP Addresses	Not Detected
Search Status	No results found
Risk Level	Low

INTERPRETATION

Since the search returned no matching events, no source IP addresses could be analyzed for suspicious authentication activity.

This indicates that:

- No repeated failed login attempts were recorded.
- No brute-force or password guessing attacks were detected.
- No suspicious source IP addresses were identified within the selected time range.
- The available Windows Security Logs did not contain relevant authentication events for analysis.

Although no suspicious IP activity was detected, this query remains useful for continuous security monitoring and can identify malicious IP addresses when failed login events are available.

SOC ANALYST INVESTIGATION

If suspicious IP addresses are detected in future searches, a SOC analyst should:

1. Identify the source IP address generating repeated failed login attempts.
2. Count the number of authentication failures from the same IP.
3. Determine the affected user accounts.
4. Check whether successful logins occurred after multiple failures.
5. Investigate the reputation of the source IP address.
6. Escalate the incident if malicious activity is suspected.

In this project, no further investigation was required because the search did not return any matching events.

SOC DASHBOARD DESIGN AND IMPLEMENTATION

A **Security Operations Center (SOC) Dashboard** is a centralized monitoring interface that provides a real-time visual representation of security events collected from various data sources. Instead of manually reviewing thousands of log entries, SOC analysts use dashboards to monitor system activities, identify suspicious events, and respond to security incidents efficiently.

In this project, a SOC Dashboard was developed using **Splunk Enterprise** to visualize Windows Security Logs. The dashboard displays authentication events, event distribution, login trends, host activity, and other security-related information through interactive charts and graphical panels. This enables analysts to quickly identify abnormal behavior and monitor the overall security status of the system.

DASHBOARD COMPONENTS

The SOC Dashboard consists of multiple visualization panels that display different aspects of security monitoring.

Panel 1 – Total Security Events

Purpose

Displays the total number of security events collected from Windows Event Logs.

SPL Query

```
index=*
```

```
| stats count
```

Visualization: Single Value

Description

This panel provides an overview of the total number of indexed security events available for analysis.

Screenshot

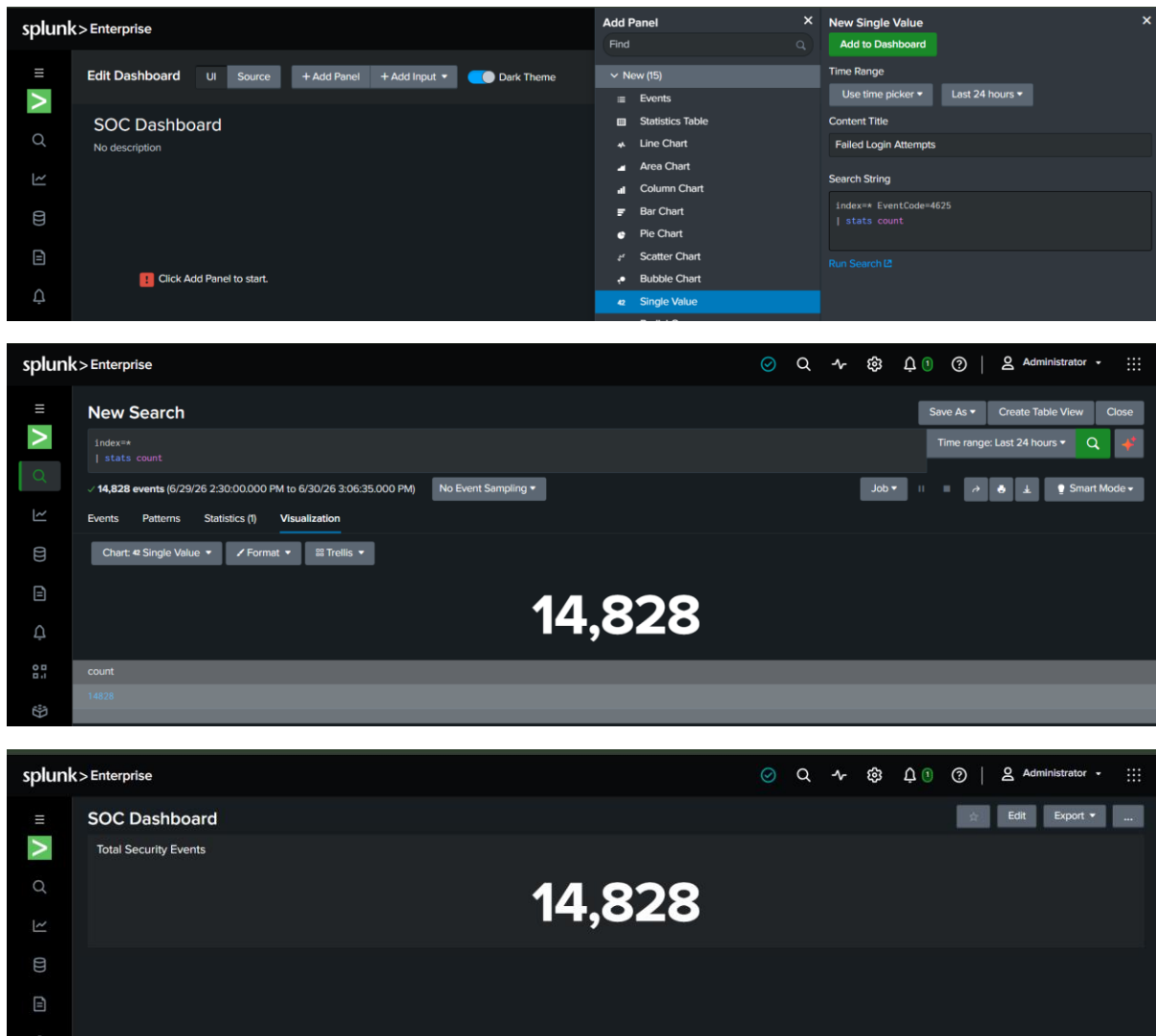


Figure 23: Total Security Events Panel

Panel 2 – Successful Login Events

Purpose

Displays the total number of successful user login events.

SPL Query

```
index=* EventCode=4624
```

```
| stats count
```

Visualization: Single Value

Description

Successful login events help analysts verify normal user authentication activities.

Screenshot

The first screenshot shows the 'SOC Dashboard' in Splunk Enterprise. The 'Add Panel' menu is open, and 'Single Value' is selected. The 'New Single Value' dialog is open, showing the search string `index=* EventCode=4624 | stats count` and the 'Add to Dashboard' button.

The second screenshot shows the 'New Search' interface. The search string `index=* EventCode=4624 | stats count` is entered. The results show 395 events. The 'Visualization' tab is selected, and the 'Chart' is set to 'Single Value'.

The third screenshot shows the 'Save Panel to Existing Dashboard' dialog. The 'SOC Dashboard' is selected. The 'Panel Title' is 'Successful Login Events'. The 'Visualization Type' is 'Single Value'.

The fourth screenshot shows the final 'SOC Dashboard' with two panels: 'Total Security Events' with a value of 14,903 and 'Successful Login Events' with a value of 395.

Figure 24: Successful Login Events Panel

Panel 3 – Failed Login Attempts

Purpose

Displays the total number of failed login attempts.

SPL Query

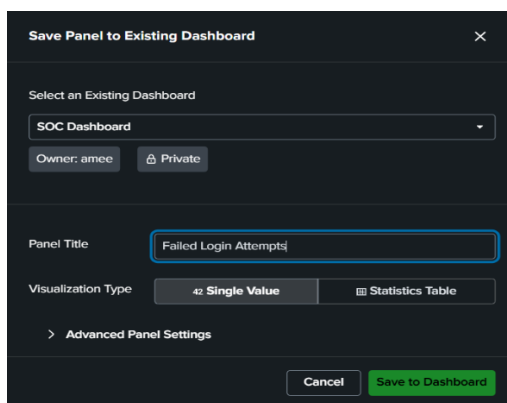
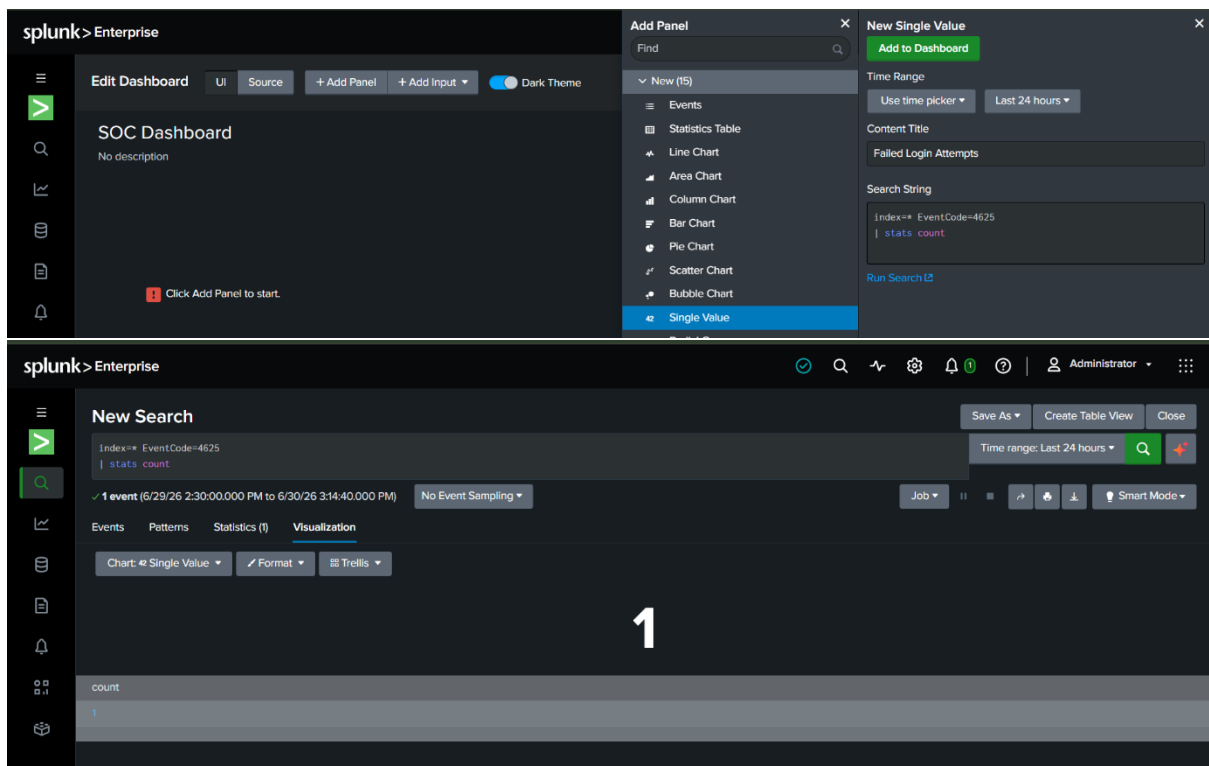
```
index=* EventCode=4625
```

```
| stats count
```

Visualization: Single Value

Description: This panel helps identify unauthorized access attempts and possible brute-force attacks.

Screenshot



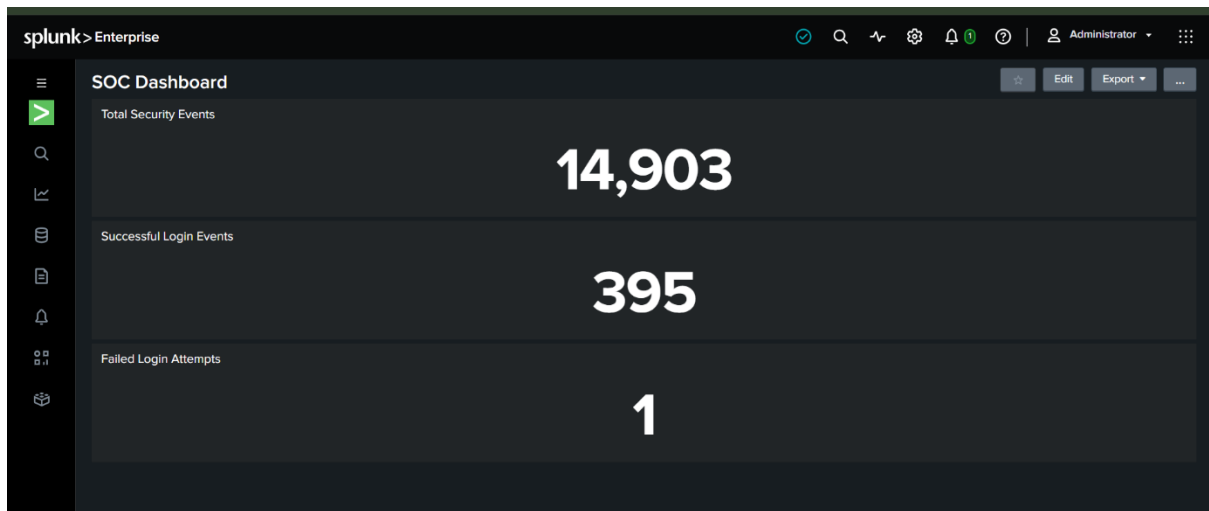


Figure 25: Failed Login Attempts Panel

Panel 4 – Event Distribution

Purpose

Displays the frequency of different Windows Event IDs.

SPL Query

`index=*`

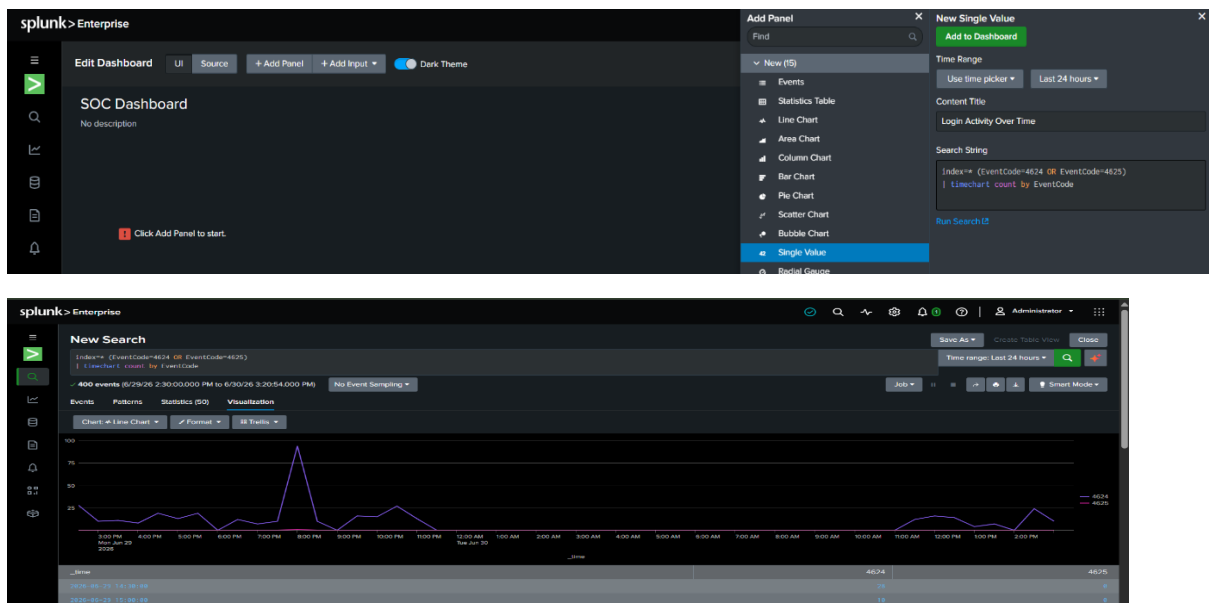
`| stats count by EventCode`

Visualization: Bar Chart or Pie Chart

Description

This visualization provides an overview of the most frequently occurring security events.

Screenshot



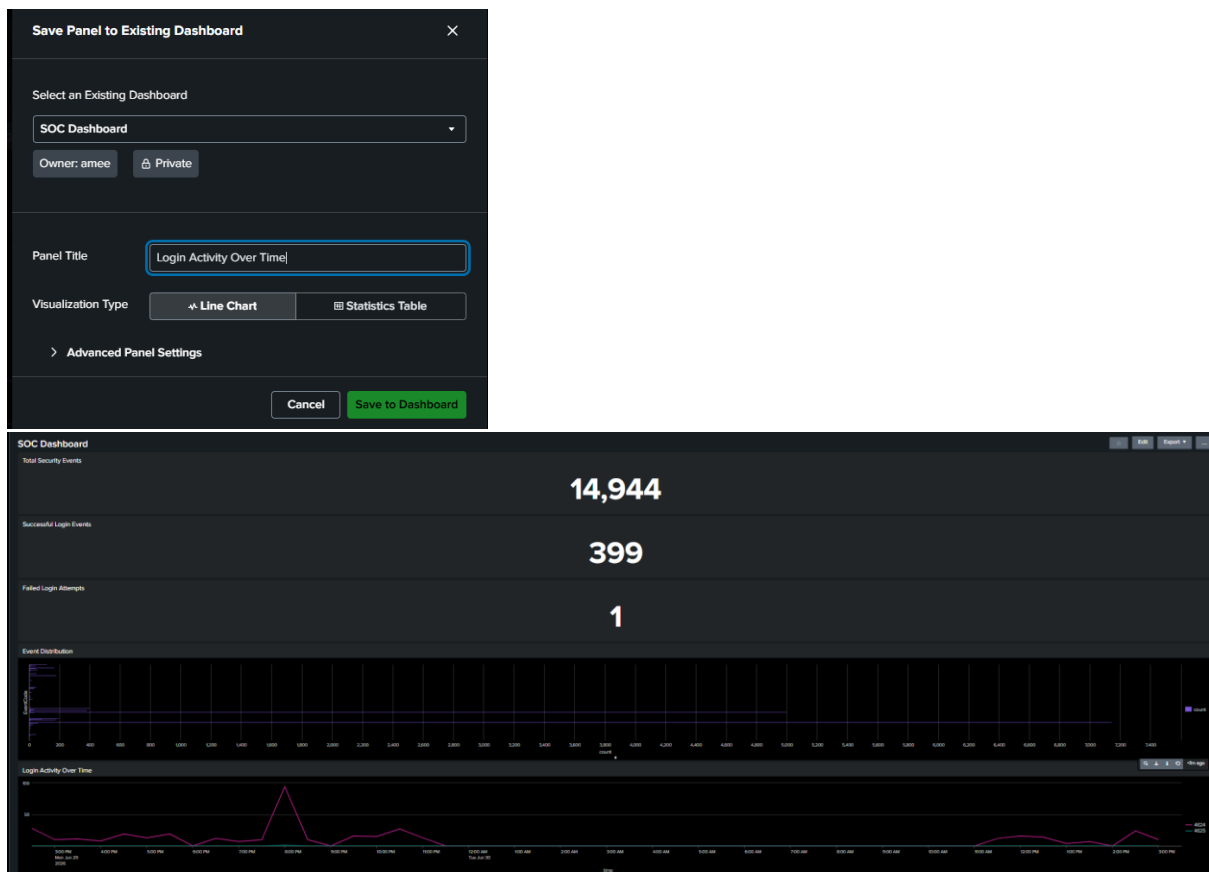


Figure 26: Event Distribution Chart

Panel 5 – Login Activity Over Time

Purpose

Shows authentication events over a selected period.

SPL Query

```
index=* (EventCode=4624 OR EventCode=4625)
```

```
| timechart count by EventCode
```

Visualization: Line Chart

Description

The timeline helps identify unusual spikes in successful or failed login attempts.

Screenshot

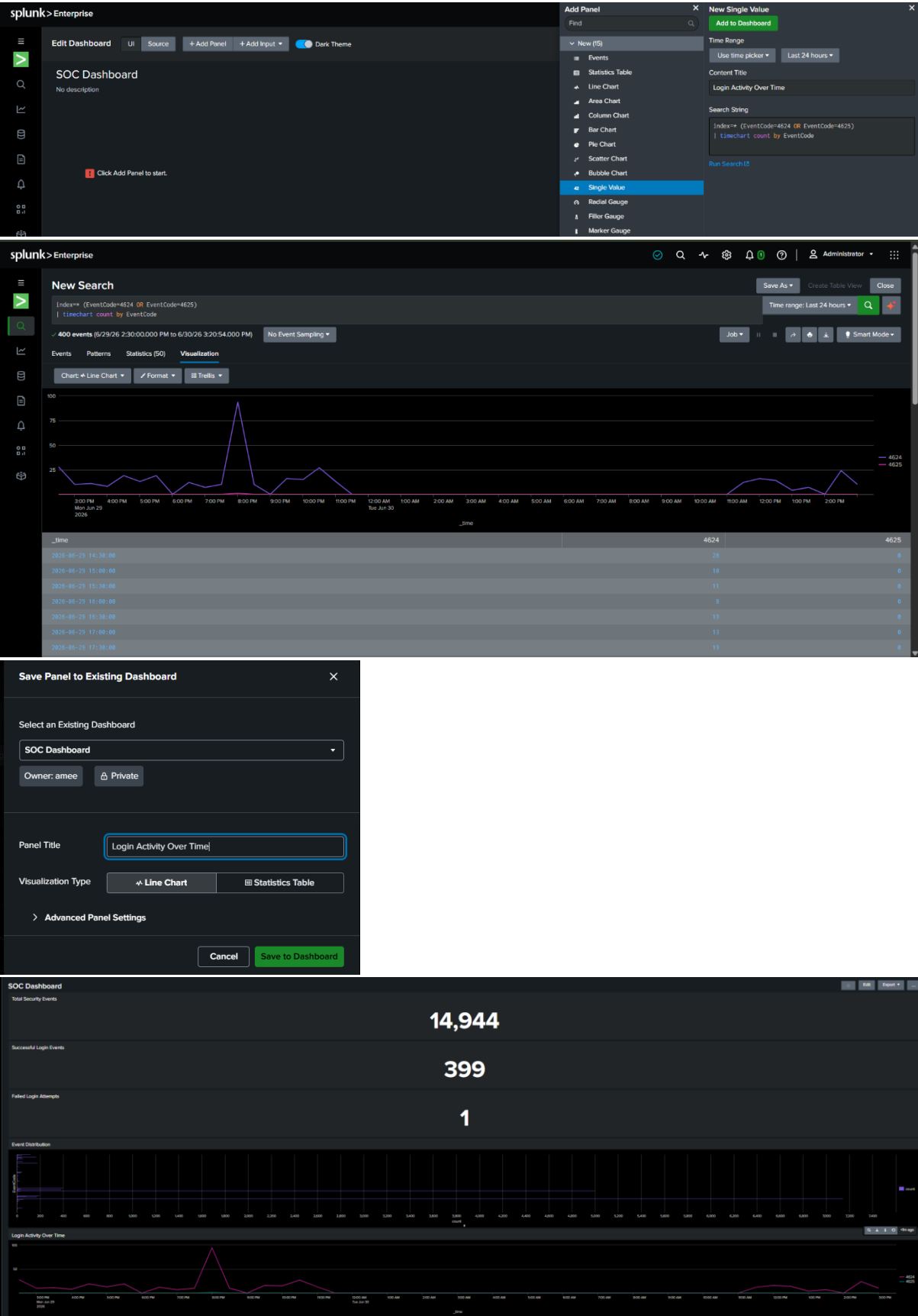


Figure 27: Login Activity Timeline

Panel 6 – Top Hosts

Purpose

Displays the systems generating the highest number of security events.

SPL Query

```
index=*
```

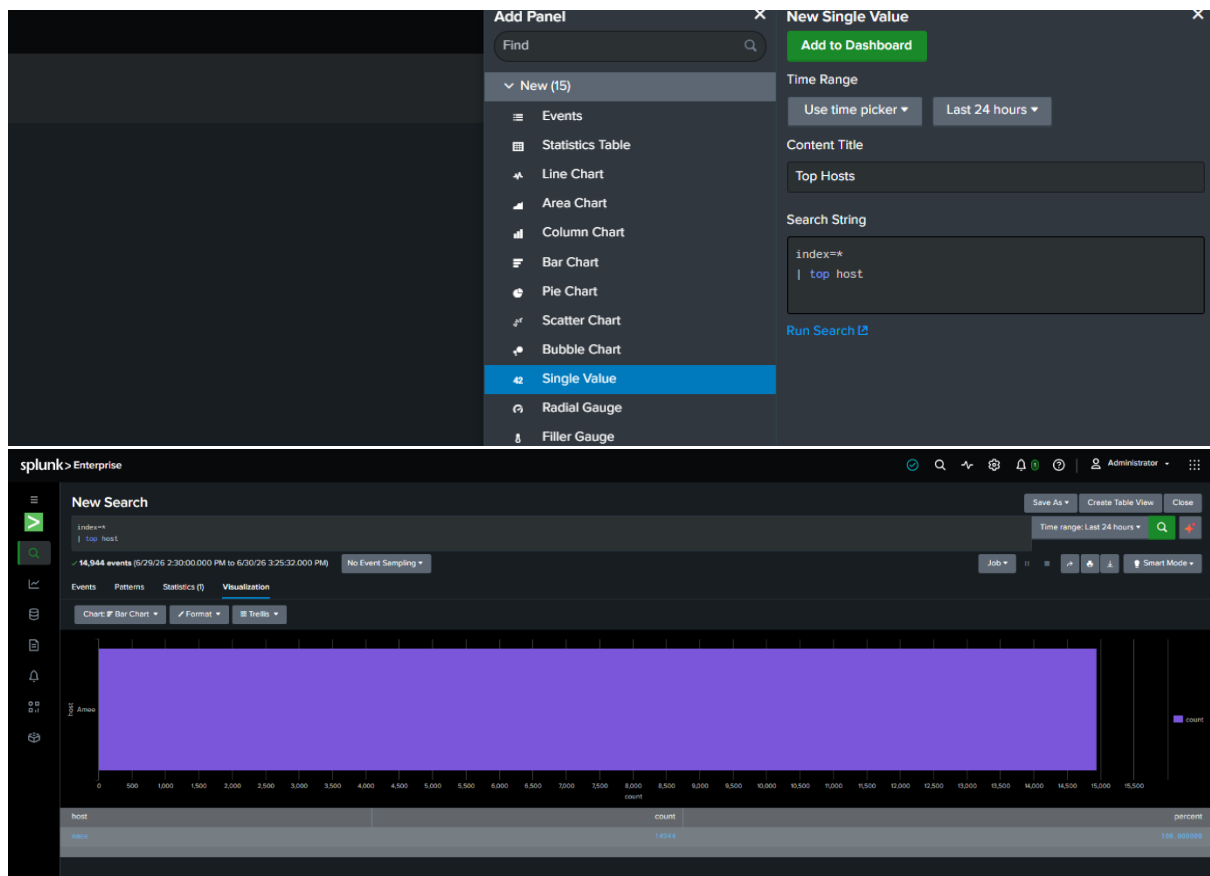
```
| top host
```

Visualization: Bar Chart

Description

This panel helps identify which hosts are producing the maximum number of security logs.

Screenshot



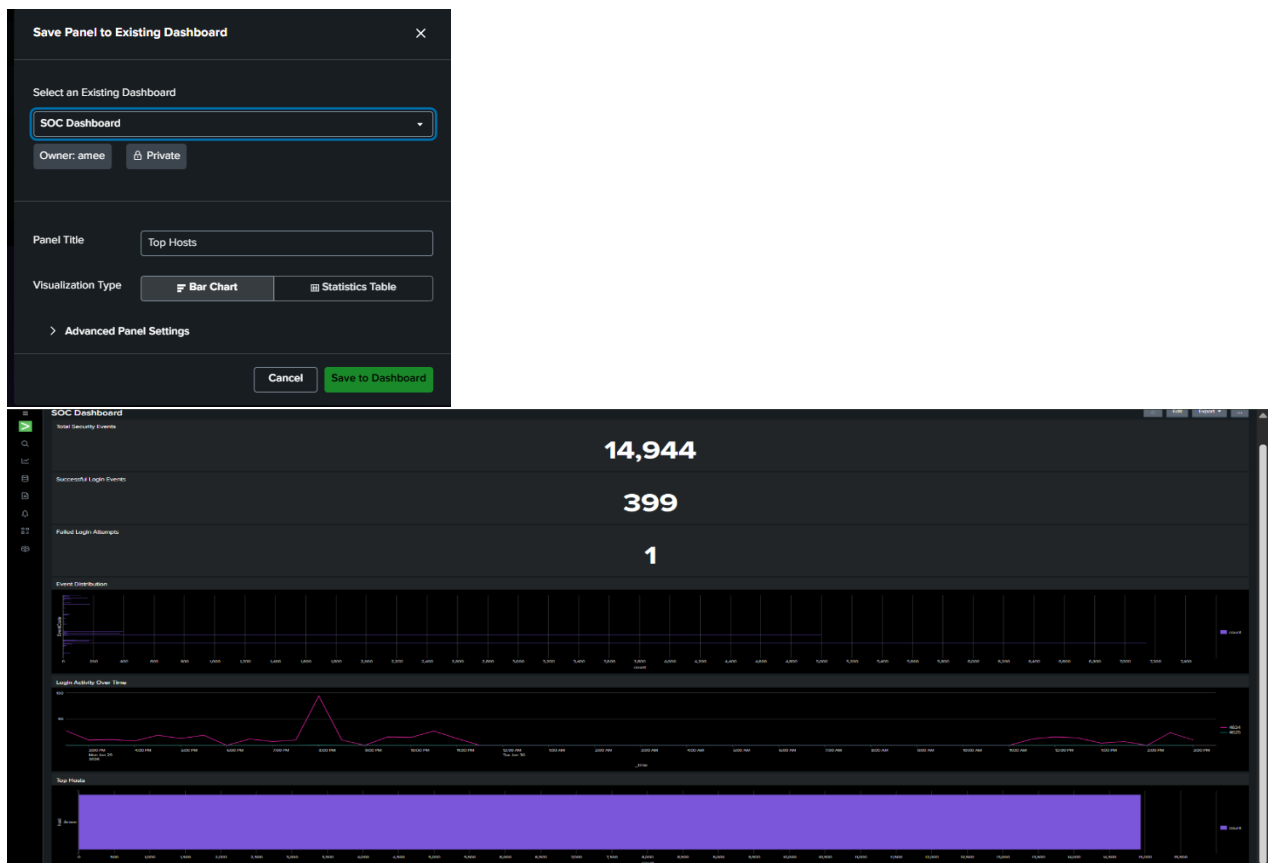


Figure 28: Top Hosts Panel

The final dashboard includes:

Screenshot

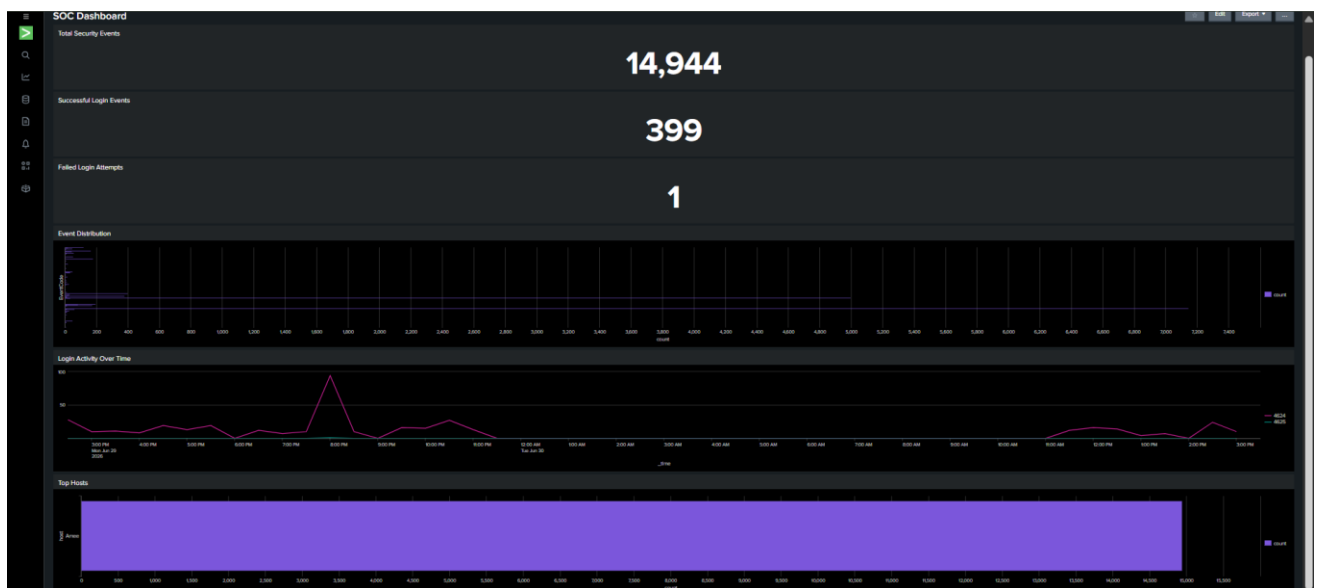


Figure 29: SOC Monitoring Dashboard

SECURITY ALERT CONFIGURATION

Security alerts are an essential component of a **Security Operations Center (SOC)** because they automatically notify analysts whenever predefined security conditions are met. Instead of continuously monitoring dashboards and logs, alerts enable security teams to detect suspicious activities in real time. These activities may include repeated failed login attempts, unusual authentication behavior, brute-force attacks, unauthorized access attempts, or other abnormal security events.

In this project, **Splunk Enterprise** was used to configure a security alert based on Windows Security Event Logs. The alert was designed to monitor **Event Code 4625 (Failed Login Attempts)** and notify the SOC analyst whenever the defined threshold was exceeded. Although no failed login events were detected during the testing period, the alert was successfully configured and is ready to detect future suspicious authentication activities.

ALERT SCENARIO

The following alert was created in Splunk Enterprise.

Alert Name: Multiple Failed Login Attempt

Purpose: Detect repeated failed login attempts that may indicate brute-force attacks or unauthorized access attempts.

Trigger Condition: Generate an alert whenever the search returns one or more matching results based on the configured SPL query.

SPL Query Used

```
index=* sourcetype="WinEventLog:Security" EventCode=4625  
| stats count by host  
| where count > 5
```

Purpose of the Query

- Searches Windows Security Logs for failed login events.
- Counts the number of failed login attempts for each host.
- Identifies hosts with more than five failed login attempts.
- Generates an alert when the specified threshold is exceeded.

STEPS TO CONFIGURE THE ALERT

Step 1: Open Splunk Enterprise and launch the **Search & Reporting** application.

Step 2: Execute the SPL query in the search bar to verify that it returns the expected events.

```
index=* sourcetype="WinEventLog:Security" EventCode=4625
```

```
| stats count by host
```

```
| where count > 5
```

The query executed successfully. During testing, no matching events were found because there were no failed login attempts recorded within the selected time range.

Screenshot

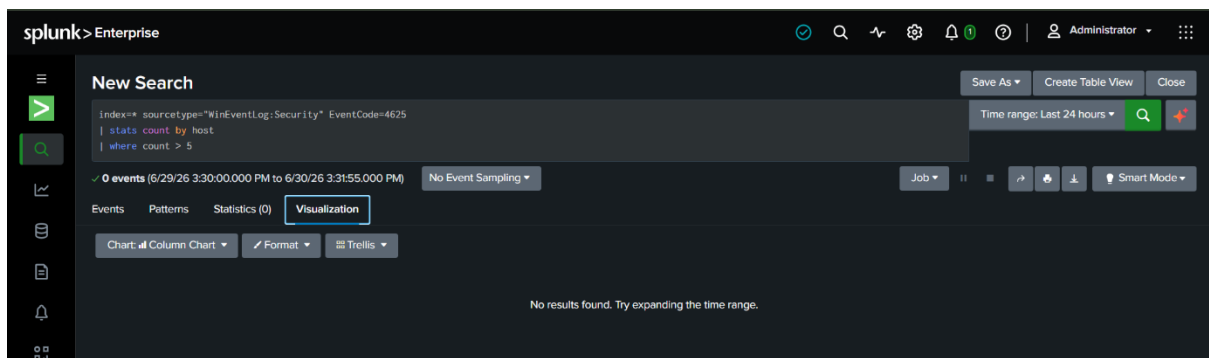


Figure 30: Executing the Alert Query

Step 3: Click **Save As** → **Alert** to create a new security alert.

Step 4: Configure the alert using the following settings.

Parameter	Value
Alert Name	Multiple Failed Login Attempt
Alert Type	Scheduled
Trigger Condition	Number of Results > 0
Schedule	Hourly (15 minutes past the hour)
Action	Add to Triggered Alerts
Status	Enabled

Screenshot

The screenshot shows the 'Save As Alert' configuration window in Splunk. The window is titled 'Save As Alert' and has a close button (X) in the top right corner. The configuration is as follows:

- Title:** Multiple Failed Login Attempts
- Description:** Optional
- Permissions:** Private (selected), Shared in App
- Alert type:** Scheduled (selected), Real-time
- Run every:** Run every hour
- At:** 15 minutes past the hour
- Expires:** 24 hour(s)
- Trigger Conditions:**
 - Trigger alert when:** Number of Results
 - is greater than:** 0
 - Trigger:** Once (selected), For each result
- Throttle:** ☐
- Trigger Actions:**
 - + Add Actions:** (button)
 - When triggered:** Add to Triggered Alerts (button), Remove (button)
 - Severity:** Medium

At the bottom of the window are 'Cancel' and 'Save' buttons.

Figure 31: Alert Configuration

Step 5: Save the alert.

After saving, Splunk continuously monitors Windows Security Logs using the configured SPL query. If future failed login events satisfy the trigger condition, the alert will automatically appear in the Triggered Alerts section.

Screenshot

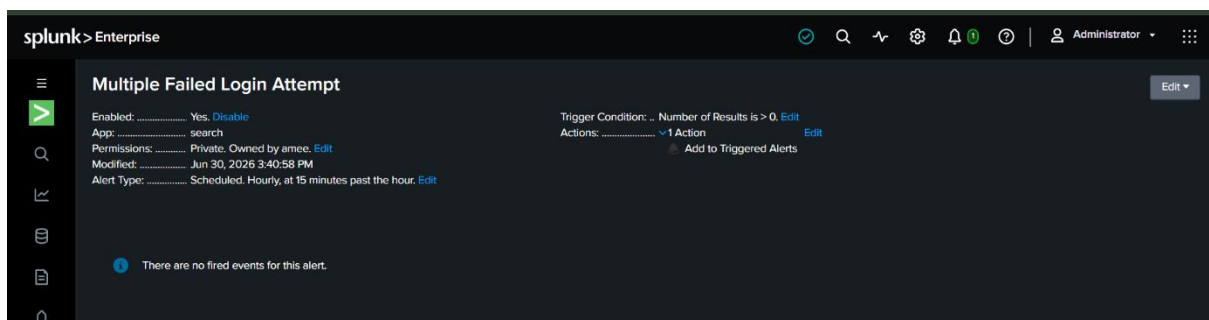


Figure 32: Alert Successfully Created

OBSERVATION

The SPL query executed successfully during testing. However, the search returned **zero matching events** because no failed login attempts (Event Code 4625) were recorded during the selected time period. Consequently, the alert did not generate any fired events.

The alert configuration page confirmed that the alert was successfully created, enabled, and scheduled to execute automatically. The message "**There are no fired events for this alert**" indicates that the monitoring rule is functioning correctly but the trigger condition was not met during testing.

Observation Table

Parameter	Observation
Alert Name	Multiple Failed Login Attempt
Alert Type	Scheduled
Alert Status	Enabled
Trigger Condition	Number of Results > 0
Query Status	Successfully Executed
Events Returned	0
Fired Alerts	None
Action Configured	Add to Triggered Alerts
Risk Level	Low

WHY IS THIS ACTIVITY SUSPICIOUS?

Multiple failed login attempts within a short period may indicate:

- Brute-force password attacks.
- Password guessing attempts.
- Unauthorized access attempts.
- Automated login scripts.
- Attempts to compromise valid user accounts.

Although occasional failed logins may occur because users enter incorrect passwords, repeated authentication failures should always be investigated by SOC analysts.

SOC ANALYST INVESTIGATION

If this alert is triggered in the future, the SOC analyst should perform the following investigation:

1. Identify the affected user account.
2. Review all failed authentication events.
3. Determine the host responsible for the failed login attempts.
4. Check whether successful logins occurred after repeated failures.
5. Review related Windows Security Events.
6. Identify any unusual authentication patterns.
7. Assess the severity of the incident.
8. Escalate the incident if malicious activity is suspected.
9. Recommend corrective actions such as password resets, account lockout, or additional monitoring.

During this project, no further investigation was required because the search returned zero matching events and no security alert was triggered.

BENEFITS OF SECURITY ALERTS

The configured security alert provides several advantages:

- Continuous monitoring of Windows Security Logs.
- Automatic detection of suspicious authentication activities.
- Faster identification of brute-force attacks.
- Reduced manual monitoring effort.
- Improved incident response capability.
- Better visibility into authentication-related security events.
- Enhanced SOC monitoring efficiency.

INCIDENT REPORT

An Incident Report is an important document prepared by Security Operations Center (SOC) analysts to record, investigate, and document security incidents detected during continuous monitoring. The report includes information about the incident, evidence collected, investigation procedures, findings, impact assessment, and recommendations. Proper documentation enables organizations to maintain security records, support compliance requirements, and improve future incident response capabilities.

In this project, Windows Security Logs were collected using Windows Event Viewer, imported into Splunk Enterprise, and analyzed using Splunk Search Processing Language (SPL). Network traffic was also examined using Wireshark to support the investigation. During the monitoring period, one failed login attempt was detected, while no evidence of unauthorized access or malicious network activity was identified.

INCIDENT SUMMARY

Parameter	Details
Incident ID	INC-001
Incident Title	Failed Login Attempt Investigation
Date of Detection	30 June 2026
Reported By	SOC Analyst
Monitoring Tool	Splunk Enterprise
Supporting Tool	Wireshark
Log Source	Windows Event Viewer
Incident Category	Authentication Failure
Severity Level	Low
Incident Status	Closed

INCIDENT DESCRIPTION

During routine monitoring of Windows Security Logs in Splunk Enterprise, authentication events were analyzed to identify failed login attempts and other suspicious activities. Dashboard analysis indicated a total of **14,944 security events**, including **399 successful login events (Event ID 4624)** and **one failed login event (Event ID 4625)**.

The failed login event was investigated to determine whether it was caused by normal user activity or a possible brute-force attack. The configured Splunk security alert monitored repeated failed login attempts; however, the alert threshold of more than five failed logins was not reached. Network traffic was also analyzed using Wireshark to verify whether any suspicious communication or unauthorized network activity was associated with the authentication event.

DETECTION METHOD

The incident was detected through manual log analysis, dashboard monitoring, and Splunk Search Processing Language (SPL) queries.

SPL Query Used

```
index=* sourcetype="WinEventLog:Security" EventCode=4625
```

The query retrieved failed login events recorded in the Windows Security Log. Dashboard panels were used to verify authentication statistics and monitor login activity over time.

Screenshots

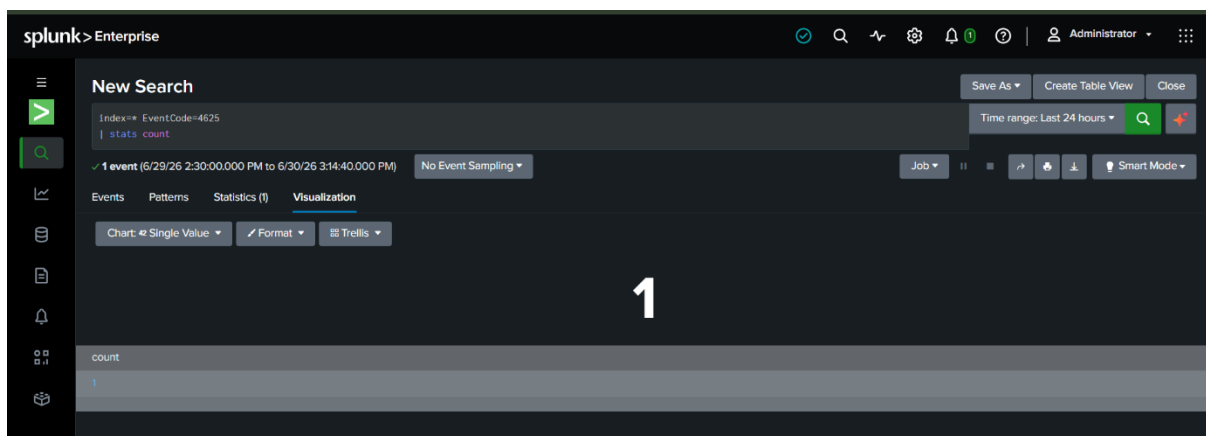


Figure 33: Failed Login Detection Query

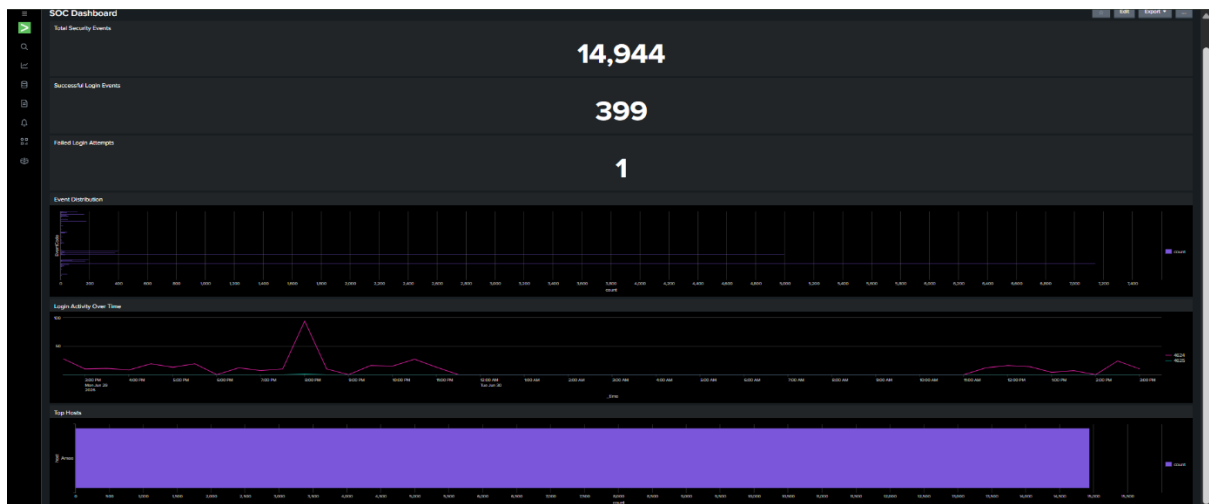


Figure 34: SOC Dashboard Showing Authentication Events

INVESTIGATION PROCESS

The following investigation activities were performed:

1. Collected Windows Security Logs using Windows Event Viewer.
2. Imported logs into Splunk Enterprise.
3. Indexed the collected logs.
4. Executed SPL queries to identify failed login attempts.
5. Reviewed authentication events using the SOC Dashboard.
6. Examined login activity over time.
7. Identified the host generating authentication events.
8. Captured and analyzed network traffic using Wireshark.
9. Correlated authentication logs with network traffic.
10. Documented the investigation findings.

EVIDENCE COLLECTED

Evidence	Description
Windows Security Logs	Authentication events collected from Event Viewer
Splunk Search Results	Failed login event identified using SPL queries
SOC Dashboard	Dashboard showing total events, successful logins, failed login attempts, event distribution, login timeline, and top host
Splunk Alert Configuration	Alert configured for multiple failed login attempts
Wireshark Capture	Network packets examined during investigation

INVESTIGATION FINDINGS

The investigation produced the following findings:

- A total of **14,944 security events** were recorded.
- **399 successful login events** were identified.
- **One failed login attempt (Event ID 4625)** was detected.
- Authentication logs were successfully collected and indexed into Splunk Enterprise.
- Login activity remained within normal operating conditions.
- The configured security alert did not trigger because the threshold of more than five failed login attempts was not reached.
- Dashboard visualizations provided clear insight into authentication activity and event distribution.
- Wireshark packet analysis showed normal network communication.
- No suspicious network connections, malware activity, or unauthorized data transfers were observed.

IMPACT ASSESSMENT

The detected failed login attempt did not result in unauthorized system access or compromise. Since only one failed authentication event was recorded and no abnormal network activity was identified, the impact on the system was assessed as Low.

Security Aspect	Assessment
Confidentiality	Not Affected
Integrity	Not Affected
Availability	Not Affected
Overall Impact	Low

CORRECTIVE ACTIONS TAKEN

The following actions were performed during the investigation:

- Reviewed the failed login event.
- Verified authentication logs.
- Examined login activity through the SOC Dashboard.
- Configured a Splunk security alert for repeated failed login attempts.
- Continued monitoring of Windows Security Logs.
- Recommended strong password policies.
- Recommended regular review of authentication logs.
- Advised continuous security monitoring using Splunk dashboards and alerts.

LESSONS LEARNED

The investigation demonstrated the importance of centralized log management and continuous monitoring using Splunk Enterprise. Dashboard visualizations made it easier to identify authentication trends, monitor login activity, and review system events. Configuring automated security alerts further improved monitoring capabilities by preparing the SOC environment to detect future brute-force attacks. Combining host-based log analysis with Wireshark network monitoring provided a more comprehensive understanding of system security.

INCIDENT REPORT SUMMARY

Activity	Observation
Total Security Events	14,944
Successful Login Events	399
Failed Login Attempts	1
Unauthorized Access	Not Detected
Suspicious Network Activity	Not Detected
Security Alert Triggered	No
Severity Level	Low
Incident Status	Closed

OUTCOME

The incident was successfully investigated using Splunk Enterprise and Wireshark. Analysis of Windows Security Logs identified **one failed login attempt**, while dashboard visualizations confirmed **14,944 total security events** and **399 successful login events**. The configured Splunk alert was successfully created but did not trigger because the defined threshold of more than five failed login attempts was not met. Wireshark analysis revealed normal network communication with no evidence of malicious activity or unauthorized access. Based on the investigation, the incident was assessed as **Low Severity**, and the case was closed after confirming that the failed login attempt did not compromise the security of the system.

THREAT ANALYSIS REPORT

A Threat Analysis Report is a structured assessment prepared by Security Operations Center (SOC) analysts to identify, evaluate, and document potential security threats observed during continuous monitoring. It helps determine possible attack vectors, assess their likelihood and impact, evaluate the overall risk level, and recommend appropriate mitigation measures.

In this project, threat analysis was performed using **Windows Event Viewer**, **Splunk Enterprise**, and **Wireshark**. Windows Security Logs were collected and analyzed in Splunk, while network traffic was monitored using Wireshark. Dashboard visualizations, search queries, and alert configurations were used to identify suspicious authentication activities and evaluate the overall security posture of the monitored system.

OBJECTIVES

The objectives of this Threat Analysis Report are:

- Identify potential security threats.
- Analyze authentication events collected from Windows Security Logs.
- Assess the likelihood and impact of identified threats.
- Determine the overall risk level.
- Recommend appropriate mitigation measures.
- Improve the effectiveness of continuous SOC monitoring.

THREAT IDENTIFICATION

During the monitoring period, the following potential threats were considered:

- Failed login attempts.
- Brute-force password attacks.
- Unauthorized access attempts.
- Account compromise.
- Suspicious network communication.
- Credential misuse by authorized users.

The SOC Dashboard recorded **14,944 security events**, including **399 successful login events (Event ID 4624)** and **one failed login attempt (Event ID 4625)**. Although one failed authentication event was observed, no evidence of repeated login failures, brute-force attacks, or unauthorized access was identified. Therefore, no critical security incident occurred during the monitoring period.

THREAT ANALYSIS TABLE

Threat	Description	Likelihood	Impact	Risk Level
Failed Login Attempt	Single failed authentication event	Low	Low	Low
Brute-Force Attack	Multiple failed password attempts	Low	High	Low
Unauthorized Access	Login using invalid or stolen credentials	Low	High	Low
Account Compromise	Misuse of valid user credentials	Low	High	Low
Suspicious Network Traffic	Unexpected communication with unknown hosts	Low	Medium	Low
Insider Threat	Misuse of authorized accounts	Low	Medium	Low

THREAT DETECTION METHODS

The following tools and techniques were used during threat analysis.

Tool	Purpose
Windows Event Viewer	Collection of Windows Security Logs
Splunk Enterprise	Log analysis, dashboards, event correlation, and security alerts
Wireshark	Network traffic capture and packet analysis
SPL Queries	Detection of failed login attempts and authentication events
SOC Dashboard	Visualization of security events and authentication trends

INDICATORS OF COMPROMISE (IOCS)

The following Indicators of Compromise (IOCs) were monitored throughout the project:

- Failed login events (Event ID 4625).
- Successful login events (Event ID 4624).
- Multiple failed login attempts.
- Successful login after repeated failures.
- Authentication activity outside normal hours.
- Repeated authentication attempts from the same host.
- Suspicious network communication.
- Unusual authentication trends.

During the monitoring period, only **one failed login attempt** was detected. No repeated authentication failures, suspicious login patterns, or malicious network communication were observed. Consequently, no significant Indicators of Compromise suggesting a successful cyberattack were identified.

THREAT SEVERITY ASSESSMENT

The identified threats were evaluated according to their likelihood of occurrence and potential impact on the monitored system.

Threat	Severity
Failed Login Attempt	Low
Brute-Force Attack	Medium
Unauthorized Access	Medium
Suspicious Network Activity	Low
Malware Communication	Not Observed
Data Exfiltration	Not Observed

Overall, the monitored environment demonstrated a **Low Risk** security posture because only one failed authentication event was recorded and no malicious activities were detected.

ANALYSIS FINDINGS

The threat analysis produced the following findings:

- Windows Security Logs were successfully collected and indexed into Splunk Enterprise.
- A total of **14,944 security events** were recorded.
- **399 successful login events** were identified.
- **One failed login attempt (Event ID 4625)** was detected.
- Dashboard visualizations successfully displayed authentication statistics, login activity over time, event distribution, and top hosts.
- The configured Splunk security alert was successfully created but did not trigger because the alert threshold (more than five failed login attempts) was not reached.
- Wireshark analysis showed normal network communication using standard protocols.
- No evidence of malware communication, unauthorized access, brute-force attacks, or data exfiltration was detected during the monitoring period.

RECOMMENDATIONS

Based on the findings of the threat analysis, the following recommendations are proposed:

- 1. Continue continuous monitoring using Splunk Enterprise dashboards.
- 2. Enable Multi-Factor Authentication (MFA) for privileged accounts.
- 3. Implement stronger password complexity requirements.
- 4. Configure additional Splunk alerts for critical Windows security events.
- 5. Monitor authentication logs regularly.
- 6. Review user accounts and login activities periodically.
- 7. Keep Windows systems and security software updated.
- 8. Perform regular vulnerability assessments.
- 9. Conduct periodic security awareness training for users.
- 10. Maintain continuous network monitoring using Wireshark or similar network analysis tools.

THREAT ANALYSIS SUMMARY

Category	Result
Windows Log Monitoring	Successfully Implemented
Authentication Monitoring	Successful
Failed Login Detection	One Event Detected
Successful Login Detection	399 Events Recorded
Dashboard Monitoring	Successfully Implemented
Security Alert Configuration	Successfully Configured
Triggered Alerts	None
Network Traffic Analysis	Completed
Critical Threats Detected	None
Overall Security Status	Stable
Overall Risk Level	Low

OUTCOME

The Threat Analysis Report demonstrated the practical implementation of Security Operations Center (SOC) monitoring using **Splunk Enterprise**, **Windows Event Viewer**, and **Wireshark**. Analysis of the collected Windows Security Logs identified **14,944 security events**, including **399 successful login events** and **one failed login attempt**. Dashboard visualizations and SPL queries enabled effective monitoring of authentication activities, while Wireshark confirmed normal network communication throughout the monitoring period.

Although one failed authentication event was detected, no evidence of brute-force attacks, unauthorized access, malware communication, or suspicious network behavior was observed. The configured Splunk security alert was successfully created and is ready to detect repeated failed login attempts in the future; however, it did not trigger because the configured threshold was not met during testing.

Overall, the monitored environment was assessed as **Low Risk**, demonstrating that centralized log collection, dashboard monitoring, automated alert configuration, and network traffic analysis provide an effective foundation for proactive SOC operations and continuous security monitoring.